Reconciling Treatment Effects Estimates across Levels of Aggregation

Paul Beaumont* Adrien Matray† Yachi Tu‡ Chenzi Xu§

June 2026

Abstract

Empirical researchers often estimate treatment effects at different levels of aggregation to address confounding shocks or to detect violations of the stable unit treatment value assumption. Using eight widely used datasets on trade, employment, syndicated loans, patents, and stock holdings, we document that disaggregated data share common properties that complicate comparisons across aggregation levels. We show that, under standard specifications, these properties can lead treatment effect estimates to differ across levels of aggregation even in the absence of shocks or SUTVA violations. These gaps arise mechanically from implicit weighting across levels of aggregation, entry and exit, and Jensen's inequality. We characterize the conditions under which they become quantitatively important and propose simple approaches that are not sensitive to these sources of bias. Using practical examples, we show how aligning estimates across levels of aggregation can help researchers separately identify confounding shocks, spillovers, and extensive-margin changes at little additional cost.

*McGill University. Email: paul.beaumont@mcgill.ca.
†Atlanta Fed. Email: adrien.matray@gmail.com.
‡UC Berkeley. Email: yaqitu@berkeley.edu.
§UC Berkeley. Email: chenzixu@berkeley.edu.

1 Introduction

Empirical researchers often estimate treatment effects at different levels of aggregation. Suppose a researcher studies the effect of a policy designed to stimulate sales for a targeted subset of firms. A natural concern in estimating the effects of the policy is that treated and control firms may be subject to different demand shocks, for instance because they operate in different product markets. With disaggregated data, the researcher typically addresses this issue by shifting the level of analysis from firm to firm-market and controlling for market-time fixed effects to absorb demand shocks ("disaggregation").

Alternatively, the researcher may be interested in how firm-level effects translate into aggregate outcomes. For instance, treated firms may expand by taking market share away from non-treated firms, leaving aggregate sales unchanged. To assess whether firm-level gains reflect reallocation or net expansion, a standard approach is to move from the firm to the industry level and regress industry-level sales growth on the share of treated firms ("aggregation"). Comparing firm-level and industry-level estimates then provides information on whether treatment effects vanish, persist or even increase (e.g., positive spillovers) once outcomes are aggregated.

Central to these approaches is the assumption that, absent market shocks or violations of the stable unit treatment value assumption (SUTVA), firm-market, firm, and industry regressions identify the same underlying estimand. Otherwise, differences across aggregation levels would be mechanical rather than economically meaningful, reflecting the fact that regressions at different aggregation levels identify distinct objects. In this paper, we show that this assumption generally fails under standard specifications and propose several ways to reconcile treatment effect estimates across levels of aggregation.

Using eight widely used datasets on trade, employment, syndicated loans, patents, and stock holdings, we first document that disaggregated data share common properties that can cause estimates to differ across levels of aggregation. First, disaggregation can substantially reweight baseline units because some units contain many more disaggregated observations than others. For instance, in Revelio–Compustat, firms at the 75th percentile have 11 times more observations when moving from the firm level to the firm–occupation level than firms at the 25th percentile. Second, entry and exit are pervasive at lower levels of aggregation, so studying changes in economic activity at a disaggregated level can drop a substantial share of economic activity. In Panjiva–Compustat, for example, moving from the firm level to the firm–destination–product level drops 82% of export activity for the median firm over a three-year horizon.

Given these properties of disaggregated data, how does changing the level of aggregation affect treatment effect estimates? We start with disaggregation. We ask under which conditions the *disaggregated* regression

$$\Delta S_{im} = \alpha_d + \beta_d \; Treated_i + \epsilon_{im} \tag{1}$$

and the *baseline* regression

$$\Delta S_i = \alpha_b + \beta_b \; Treated_i + \epsilon_i \tag{2}$$

identify the same estimand, that is, $\beta_d = \beta_b$. For illustration, we take the firm-market as the disaggregated level, the firm as the baseline level, and let S denote sales. In this setting, disaggregation allows ruling out product demand shocks (e.g., Berman, Martin and Mayer, 2012; Paravisini, Rappoport, Schnabl and Wolfenzon, 2014). In practice, S can represent any variable whose value at the baseline level can be exactly decomposed as the sum of its values at the disaggregated level. For instance, moving from firm-level bank credit to firm-bank-level credit allows absorbing credit demand shocks (e.g., Khwaja and Mian, 2008; Iyer, Peydró, da Rocha-Lopes and Schoar, 2014), while moving from area-level employment to industry-area employment allows absorbing industry-level shocks (e.g., Mian and Sufi, 2014; Fonseca and Matray, 2024).

We show that under most standard empirical specifications, the disaggregated and baseline regressions identify different estimands, i.e., $\beta_d \neq \beta_b$. There are three main reasons for the discrepancy. First, when observations are equally weighted, firms active in more markets receive greater weight in the firm-market regression, leading to a different treatment effect estimate. Second, even with appropriate weighting, Jensen's inequality implies that the firm-level growth rate and the average firm-market-level growth rate will differ when growth rates are non-linear in sales levels (e.g., log growth rate). Third, firms may enter or exit markets, generating zeros at the firm-market level and, consequently, missing values when using standard or log growth rates.

We propose two approaches that ensure the disaggregated and baseline regressions identify the same estimand, $\beta_d = \beta_b$ ("disaggregation invariance"). First, using the midpoint growth rate $\Delta S_t = 2 \cdot (S_t - S_{t-h})/(S_t + S_{t-h})$ and weighting firm-market observations by $(S_{imt} + S_{imt-h})/(S_{it} + S_{it-h})$ ("midpoint weights") yields identical treatment effect estimates across aggregation levels. Second, estimating treatment effects on sales levels using Poisson regressions also delivers disaggregation invariance. Both approaches accommodate zeros at the disaggregated level, which is central to restoring disaggregation invariance. Midpoint weights align firm–market observations with their contribution to firm-level growth. Poisson estimation instead exploits the fact that sales levels aggregate linearly: with firm–market fixed effects in the disaggregated regression, the first-order condition for the treatment effect aggregates to the corresponding firm-level first-order condition.

We proceed with simulations to quantify the gap between firm- and firm-market estimates under standard empirical specifications. Our data-generating process builds on Eaton, Kortum and Kramarz (2011) and is sufficiently flexible to accommodate different growth-rate transformations and levels of aggregation while remaining analytically tractable. This allows us to derive the implied average treatment effect under each specification and assess precisely when empirical estimates diverge from their population counterparts. Market entry and sales conditional on entry depend on firm productivity, and we model treatment as a positive productivity shock.

Using simulations, we characterize when disaggregation invariance breaks under standard empirical specifications. Implicit weights become important when treatment effects are heterogeneous across firms and correlate with the number of markets in which firms operate. Jensen's inequality, in turn, matters when treatment affects the within-firm dispersion of growth rates. Finally, entry and exit only cause disaggregation invariance to break when treatment affects market participation. We then show that midpoint growth rates with midpoint weights preserve disaggregation invariance across all three settings. Moreover, this approach delivers an exact decomposition of firm-level growth into intensive-margin, entry, and exit components, allowing researchers to quantify directly the contribution of each channel to the overall treatment effect.

We then show how disaggregation invariance can be used to detect confounding shocks. We introduce market-level shocks that are correlated with treatment status and compare firm- and firm-market estimates with and without market fixed effects. In our simulation, standard log-growth specifications generate firm- and firm-market coefficients that appear nearly identical even when confounding shocks are large, because the downward bias from omitted market shocks happens to be of similar magnitude to the downward bias from omitted entry-and-exit adjustments. In contrast, midpoint growth rates preserve comparability across aggregation levels, allowing researchers to isolate the contribution of market shocks and directly measure the resulting bias in firm-level estimates.

We then turn to aggregation. Let $Share\ Treated_j$ denote the share of treated firms in j. We say that aggregation invariance holds if, in the absence of violations of SUTVA, the *aggregated* regression

$$\Delta S_j = \alpha_a + \beta_a \, Share\ Treated_j + \epsilon_j \tag{3}$$

identifies the same estimand as the firm-level regression, so that $\beta_a = \beta_b$. For illustration, we take j to denote the industry. Comparing firm- and industry-level coefficients therefore aims to detect SUTVA violations across firms within industries (e.g., Garthwaite, 2014; Helm, 2020). Researchers also aggregate firm-level outcomes to the geographical-area level to detect spillovers across firms within areas (e.g., Kline and Moretti, 2014; Huber, 2018; Giroud and Mueller, 2019; Mian, Sarto and Sufi, 2023), and aggregate firm-bank outcomes to the firm level to study credit reallocation across banks within firms (e.g., Khwaja and Mian, 2008; Cingano, Manaresi and Sette, 2016; Beck, Da-Rocha-Lopes and Silva, 2021).

Aggregation is distinct from disaggregation in that it changes the regressor, not only the level of observation. Disaggregation relies on the same treatment indicator across aggregation levels, so disaggregation invariance can be restored by appropriately specifying the method of aggregation. Aggregation, by contrast, replaces the firm-level treatment indicator with an industry-level share of treated firms. As a result, firm- and industry-level regressions rely on different population moment conditions, and the firm-level condition does not in general imply its industry-level counterpart, even absent spillovers. In addition, aggregation invariance breaks for the same reasons as disaggregation invariance: firm entry and exit, implicit weighting, and Jensen's inequality.

We first show that the two approaches proposed to restore disaggregation invariance do not, in general, deliver aggregation invariance. Poisson regressions fail because the conditional mean at the industry level is not a linear function of the industry-level share of treated firms. Midpoint growth rates fail because aggregation invariance would require weighting the industry-level share of treated firms by midpoint weights that are themselves affected by treatment, generating a simultaneity bias.

We achieve aggregation invariance in two steps. First, abstracting from the extensive margin, we show that the firm- and industry-level population equations can be aligned by including the industry-level share of treated firms in the firm-level regression. We argue that this specification has a straightforward statistical and economic interpretation. From a statistical perspective, the coefficient on $Share\ Treated_j$ captures the wedge between firm- and industry-level estimates. Under our aligned specifications, this coefficient is exactly zero if and only if $\hat{\beta}_b$ and $\hat{\beta}_a$ coincide, making it straightforward to assess whether differences across aggregation levels are statistically significant. From an economic perspective, the firm-level regression tests whether exposure to treated firms within the industry affects firm growth beyond a firm's own treatment status. This "exposure to treated" test is often used as an alternative to aggregation to detect SUTVA violations (Crépon, Duflo, Gurgand, Rathelot and Zamora, 2013; Cai and Szeidl, 2024; Lerche, 2025).[1] Our approach shows that the "exposure to treated" and aggregation approaches are equivalent when aggregation invariance holds.

Second, we bridge firm- and industry-level regressions by exploiting a decomposition of industry growth into intensive- and extensive-margin components. Weighting firm-level observations by lagged industry market shares allows us to recover the treatment effect on incumbent firms under the standard growth rate. We then recover the total industry-level effect by separately estimating the contribution of firm entry through a regression of the entry rate on the industry share of treated firms. This two-step approach also reconciles inference across aggregation levels when standard errors are clustered at the aggregate level.

Using simulations, we show that standard methodologies can generate misleading comparisons between firm- and industry-level estimates even in the absence of SUTVA violations. First,

[1] This specification is sometimes referred to as the "linear-in-means" model in the peer effects literature (Manski, 1993).

industry-level regressions implicitly place greater weight on larger firms, so aggregation invariance breaks when industries are concentrated and idiosyncratic shocks to dominant firms become quantitatively important. Second, under log growth rates, aggregation invariance also breaks through Jensen's inequality because treatment exposure is mechanically linked to the within-industry dispersion of firm growth rates.

Finally, we show how our methodology can be used to detect and decompose spillover effects. We consider a setting in which treatment simultaneously stimulates firm entry while generating negative spillovers on incumbent firms. Under standard log-growth specifications, firm- and industry-level coefficients appear very similar because the positive contribution of entry offsets the negative effect on incumbent firms in aggregate industry sales growth. As a result, comparing coefficients across aggregation levels alone may incorrectly suggest the absence of spillovers. In contrast, our two-step methodology separately identifies intensive- and extensive-margin effects, making it possible to distinguish spillovers operating through incumbent firms from those operating through entry and to recover the full aggregate effect of treatment.

The remainder of the paper proceeds as follows. Section 2 documents two properties of disaggregated data across eight datasets. Section 3 studies disaggregation: it defines disaggregation invariance, shows why standard specifications break it, and establishes that midpoint growth rates with midpoint weights and Poisson estimation restore it. Section 4 turns to aggregation, where restoring invariance requires aligning the firm- and industry-level population equations and a two-step decomposition that separates incumbent and entry effects, before the paper concludes.

Contribution to the literature. This paper contributes to a growing methodological literature on the disaggregation and aggregation of reduced-form empirical estimates.

Research on disaggregation was spurred by the increasing availability of detailed microdata, which makes it possible to link aggregate changes to micro-level components. Davis and Haltiwanger (1992) first shows how aggregate employment movements can be decomposed into establishment-level contributions and introduce the midpoint growth rate to accommodate zeroes arising from entry and exit, thereby delivering exact decompositions of aggregate changes. Similar methodologies have been applied in a wide range of settings to trace the micro origins of aggregate moments.[2]

By contrast, relatively less attention has been paid to whether and how changes in aggregation levels affect treatment effect estimates. Notable exceptions include Moulton (1990), who discusses inference issues when regressing disaggregate outcomes on aggregate ones. Related to the treatment of zeroes in disaggregated data, Cohn, Liu and Wardlaw (2022) and Chen and Roth (2024) show

[2] See e.g., Baily, Hulten, Campbell, Bresnahan and Caves (1992); Olley and Pakes (1996); Foster, Haltiwanger and Krizan (2001); Hsieh and Klenow (2009); Gabaix (2011); Di Giovanni, Levchenko and Mejean (2014, 2017); Fitzgerald and Haller (2018); Boualam and Mazet-Sonilhac (2025).

that common transformations used to deal with zeroes can yield reduced-form estimates that are biased or lack a clear economic interpretation.[3] Degryse, De Jonghe, Jakovljević, Mulier and Schepens (2019) shows that the use of high-dimensional fixed effects in disaggregated data can induce selection and bias reduced-form estimates. Finally, Beaumont, Libert and Hurlin (2019) shows that using midpoint growth rates with weighted least squares facilitates the incorporation of the extensive margin in the joint estimation of bank and firm fixed effects (Amiti and Weinstein, 2018), an approach subsequently adapted by Fonseca and Matray (2024) and Matray, Müller, Xu and Kabir (2024) to obtain estimates that are consistent across levels of disaggregation. Our paper contributes to this strand of research by formally characterizing the methodological issues raised by estimating treatment effects with disaggregated data, illustrating when discrepancies across levels of disaggregation arise, and proposing simple solutions to restore disaggregation invariance.

Faced with an increasing availability of micro-level estimates of causal responses (Angrist and Pischke, 2010), a substantial body of work, summarized in Nakamura and Steinsson (2018), has developed methods to map reduced-form estimates into aggregate outcomes. Our paper relates specifically to the common practice of comparing estimates across aggregation levels to infer general equilibrium or spillover effects (e.g., Khwaja and Mian, 2008; Kline and Moretti, 2014; Huber, 2018; Giroud and Mueller, 2019).[4] In this spirit, Mian, Sarto and Sufi (2023) contrasts within- and across-region estimates to isolate regional amplification mechanisms of credit shocks, while Herreno (2023) uses a structural framework to map relative cross-sectional estimates of bank credit shocks into aggregate elasticities. Relatedly, Ferracci, Jolivet and van den Berg (2014) develops a two-step framework that identifies direct and indirect treatment effects by exploiting variation in individual treatment status and in market-level treatment shares. At the disaggregated level, Berg, Reisinger and Streitz (2021) shows that ignoring spillovers can severely bias treatment effect estimates, while Huber (2023) provides econometric guidance for estimating spillover effects in the presence of nonlinearities, measurement error, and multiple spillover channels. Chemla and Hennessy (2026) shows how moral hazard can undermine standard tests for SUTVA violations when firms gather information before deciding whether to take up a treatment. In contrast, we show that, even in settings without SUTVA violations or additional frictions, treatment effect estimates generally differ across aggregation levels under standard specifications, casting doubt on the ability of aggregation-based SUTVA diagnostics to avoid false positives. We characterize the

[3]Silva and Tenreyro (2006) shows that, in the presence of heteroskedasticity, OLS estimates of log-linearized models are necessarily biased due to Jensen's inequality and recommend Poisson specifications instead.

[4]Alternative approaches include deriving conditions under which cross-sectional estimates can inform general equilibrium effects (e.g., Chodorow-Reich, 2019; McKay and Wolf, 2023; Wolf, 2023), combining multiple cross-sectional estimates to recover aggregate effects (e.g., Arkolakis, Costinot and Rodríguez-Clare, 2012; Guren, McKay, Nakamura and Steinsson, 2021; Sraer and Thesmar, 2023), or constructing general equilibrium models that target reduced-form moments to assess aggregate effects (e.g., Chodorow-Reich, 2014; Catherine, Chaney, Huang, Sraer and Thesmar, 2022).

sources of these discrepancies across aggregation levels and propose simple approaches to restore aggregation invariance.

Finally, our paper builds on a growing literature that seeks to provide actionable solutions to underappreciated econometric challenges (e.g., Bertrand, Duflo and Mullainathan, 2004; Petersen, 2008; Gormley and Matsa, 2013; Jiang, 2017; Chodorow-Reich, 2019; Goldsmith-Pinkham, Sorkin and Swift, 2020; Borusyak, Hull and Jaravel, 2021; Callaway and Sant'Anna, 2021; Baker, Larcker and Wang, 2022; Cohn, Liu and Wardlaw, 2022; Lerner and Seru, 2022; Borusyak and Hull, 2023; Chen and Roth, 2024). We contribute to this literature by introducing the notion of disaggregation and aggregation invariance, which formalizes the common intuition that, in the absence of compositional or general equilibrium effects, treatment effects should not depend on the level of aggregation. We show, using simulations and real data, that traditional methods do not deliver disaggregation or aggregation invariance, highlighting the need to revisit common rules of thumb in applied work.

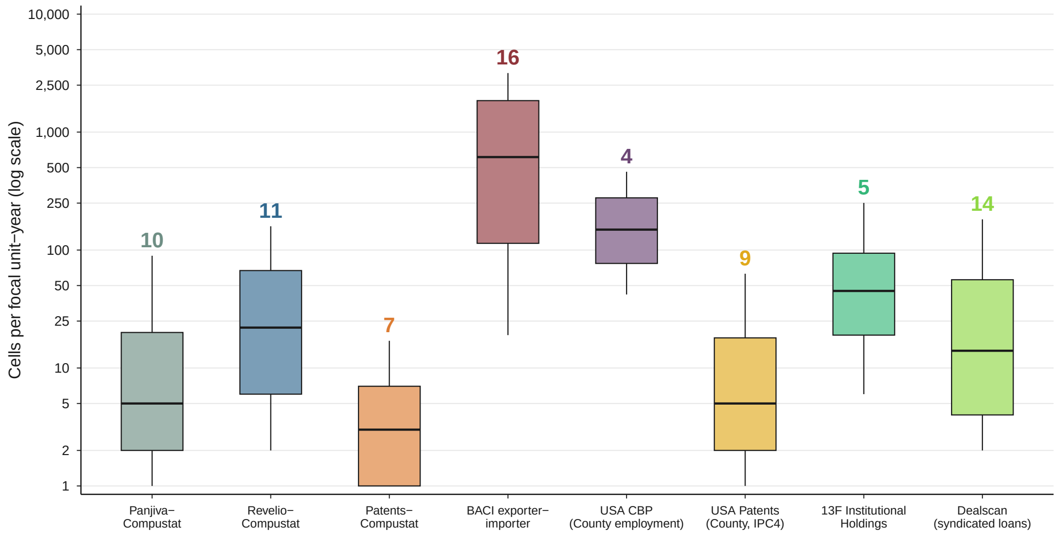
2 Descriptive statistics on disaggregated data

To what extent can changing the level of aggregation change the object being estimated? We address this question by documenting two general properties of disaggregated datasets. The analysis draws on several disaggregated datasets widely used in economics and finance: (i) U.S. export data from Panjiva–Compustat; (ii) U.S. employment data from Revelio–Compustat; (iii) patent data linked to Compustat firms; (iv) the BACI bilateral trade database; (v) U.S. county-level employment data from the U.S. Census Bureau; (vi) U.S. patent data at the county level; (vii) syndicated loan data from Dealscan; and (viii) institutional investor stock holdings from 13F filings. For each dataset, we define a baseline level and a disaggregated level, with the latter nested within the former. We impose two common restrictions across datasets. First, baseline units must be observed in at least two periods, so that the analysis focuses on continuing units. Second, observations with missing or non-positive values ("inactive observations") are dropped before aggregation. Appendix A describes the levels used in each dataset and the corresponding construction steps.

A first property of disaggregated datasets is that baseline units differ in the number of observations they contain at the disaggregated level. As a result, changing the level of observation changes the weight placed on different baseline units and can therefore change the estimand. Figure 1 reports the distribution of the number of disaggregated observations per baseline unit across datasets. The figure also reports the ratio of the 75th percentile to the 25th percentile above each box.

We find substantial variation across baseline units in the number of observations at the disaggregated level. In the Panjiva–Compustat data, for example, a firm at the 75th percentile has positive exports in 10 times more destination–HS6 product observations than a firm at the 25th

Figure 1: Number of disaggregated observations per baseline unit across datasets



Note: The figure reports the distribution of the number of disaggregated observations per baseline unit. We compute this number separately by year and then pool baseline observations across years. The middle line denotes the median, boxes span the 25th to the 75th percentile, and whiskers span the 10th to the 90th percentile. The number above each box indicates how many more disaggregated observations a baseline unit at the 75th percentile has relative to a baseline unit at the 25th percentile. In the Panjiva–Compustat data, for example, a firm at the 75th percentile has positive exports in 10 times more destination–HS6 product observations than a firm at the 25th percentile. The aggregation levels are defined as follows, with the baseline level listed first and the disaggregated level listed second: (i) *Panjiva–Compustat:* firm and firm × destination × product (HS6); (ii) *Revelio–Compustat:* firm and firm × occupation; (iii) *Patents–Compustat:* firm and firm × technology class; (iv) *BACI bilateral trade data:* exporter–importer pair and exporter–importer pair × product (HS6); (v) *U.S. county-level employment data:* county and county × NAICS6 industry; (vi) *U.S. patent data:* county and county × IPC4 technology class; (vii) *13F institutional holdings:* fund manager and fund manager × stock; and (viii) *Dealscan:* bank and bank × SIC4 industry.

percentile. Similar patterns appear in other firm-level datasets: firms differ substantially in the number of occupations or patent classes they contain. The gap is especially pronounced in BACI, where exporter–importer pairs differ sharply in the number of products traded, and in Dealscan, where banks differ substantially in their number of syndicated loan borrowers. Dispersion is smaller but still meaningful in more aggregated datasets such as U.S. county-level employment and 13F institutional holdings. Overall, the evidence shows that a cell-level specification mechanically assigns more weight to baseline units with many active disaggregated observations. Without explicit reweighting, changing the level of observation therefore changes the relative importance of baseline units in the estimated object.

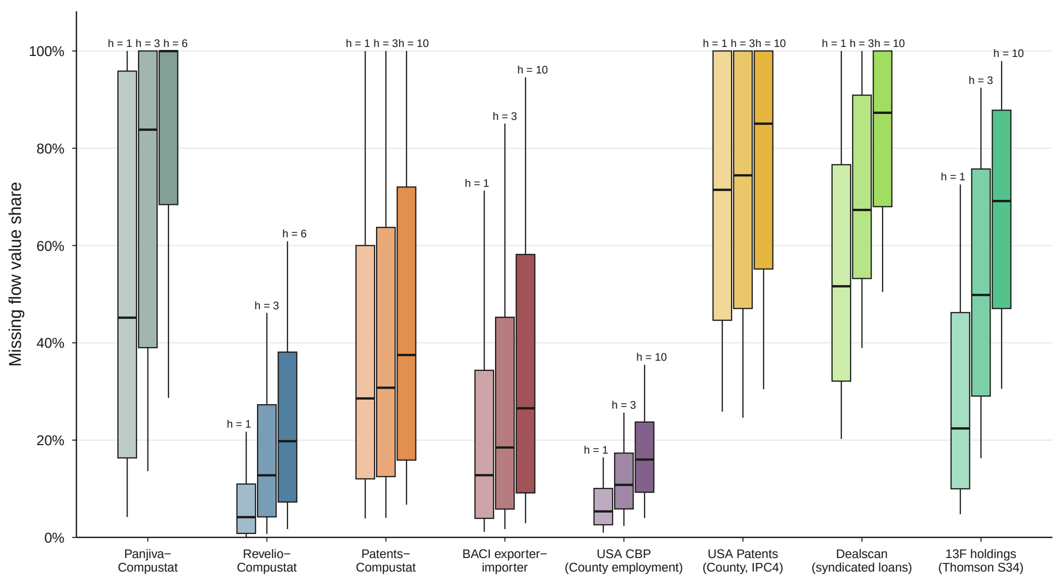
A second property of disaggregated datasets is the pervasive importance of entry and exit at lower levels of aggregation. Even when a growth rate is well defined at the baseline level, it may be undefined at the disaggregated level because some observations are nonzero in only one of the two periods. As a result, changing the level of aggregation can change the estimand mechanically: growth rates that require positive values in both periods drop entry and exit observations, which breaks the link between growth rates across levels of aggregation.

We illustrate the magnitude of this problem by measuring how much baseline-unit activity is lost when entry or exit observations are dropped. Specifically, we define the "extensive-margin share" as

$$\text{Extensive margin share}_{i,t} = \frac{\text{Entry value}_{i,t} + \text{Exit value}_{i,t-h}}{S_{i,t} + S_{i,t-h}}. \tag{4}$$

A value of 30% means that a specification requiring positive disaggregated observations in both periods drops 30% of the baseline unit's activity in value. We restrict attention to baseline units with $S_{i,t-h} > 0$ and $S_{i,t} > 0$, so that the baseline unit is present in both periods, where h denotes the time horizon. The entry value is the value at t of disaggregated observations that are inactive at $t-h$ but active at t. The exit value is the value at $t-h$ of disaggregated observations that are active at $t-h$ but inactive at t.

Figure 2: Extensive margin share across datasets



Note: The figure reports the distribution of the extensive margin share, as defined by Equation 4, across datasets. A value of 30% means that a specification requiring positive disaggregated observations at the beginning and end of the period drops 30% of the baseline unit's activity in value. The middle line denotes the median, boxes span the 25th to the 75th percentile, and whiskers span the 10th to the 90th percentile. We compute the extensive margin share for a short, medium, and long horizon h. The short horizon is one year, the medium horizon is three years, and the long horizon is ten years when possible, or six years otherwise. We compute the shares separately by year and then pool baseline observations across years. The aggregation levels are defined as follows, with the baseline level listed first and the disaggregated level listed second: (i) *Panjiva–Compustat:* firm and firm × destination × product (HS6); (ii) *Revelio–Compustat:* firm and firm × occupation; (iii) *Patents–Compustat:* firm and firm × technology class; (iv) *BACI bilateral trade data:* exporter–importer pair and exporter–importer pair × product (HS6); (v) *U.S. county-level employment data:* county and county × NAICS6 industry; (vi) *U.S. patent data:* county and county × IPC4 technology class; (vii) *13F institutional holdings:* fund manager and fund manager × stock; and (viii) *Dealscan:* bank and bank × SIC4 industry.

Figure 2 reports the results. Three patterns emerge. First, even over short horizons, disaggregation can lead researchers to drop a substantial share of a baseline unit's activity. In Panjiva–Compustat, for example, the median extensive-margin share is 45% at the one-year horizon, with a 75th percentile above 95%. Second, the importance of the extensive margin depends on the nature of the underlying economic activity. Entry and exit are more important, for instance, in stock portfolios or firm-level export flows than in firms' workforce composition or countries' product trade flows. Third, the extensive margin becomes more important as the horizon increases. In Revelio–Compustat, the median extensive-margin share is 5% at the one-year horizon but reaches 20% at the six-year horizon. Longer differences give disaggregated cells more time to enter or exit, so specifications based only on continuing cells increasingly restrict attention to a selected subset of the baseline unit's activity.

Our analysis suggests that disaggregated datasets have specific properties that can cause average treatment effect estimates to substantially differ across levels of aggregation. In the next two sections, we formally study when, and to what extent, average treatment effect estimates diverge across levels, focusing first on disaggregation and then on aggregation.

3 Disaggregation

3.1 Disaggregation invariance

We now formalize the notion of disaggregation invariance. The baseline outcome is firm-level sales, S_{it}, which can be exactly decomposed into firm–market components:

$$S_{it} = \sum_m S_{imt}.$$

Throughout the paper, we consider three commonly used growth-rate definitions: log growth, standard growth, and midpoint growth. For any horizon $h \geq 1$, these growth rates are defined as

$$\Delta^l S_t = \log S_t - \log S_{t-h}, \qquad \Delta^s S_t = \frac{S_t - S_{t-h}}{S_{t-h}}, \qquad \Delta^{mp} S_t = 2\,\frac{S_t - S_{t-h}}{S_t + S_{t-h}}.$$

We omit the horizon h from the growth operators to simplify notation. Disaggregation consists in replacing $\{S_{it}\}_i$ by $\{S_{imt}\}_{i,m}$ in equation (2). We say that a specification achieves disaggregation invariance if the disaggregated and baseline regressions target the same estimand, i.e., $\beta_b = \beta_d$. Our baseline and disaggregated specifications (1) and (2) both imply

$$\beta = E(\Delta S \mid Treated = 1) - E(\Delta S \mid Treated = 0). \tag{5}$$

Therefore, in the following, we study under which conditions the conditional mean $E(\Delta S \mid Treated = 1)$ is invariant to the level of aggregation. Since the same reasoning applies to control firms, the invariance of β to disaggregation follows immediately by differencing.

Importantly, our definition of disaggregation invariance does not require any assumption on treatment being randomly assigned at the firm or firm–market level. In practice, researchers often disaggregate precisely because they suspect that treatment is not random at the firm level, for instance because treated and control firms operate in systematically different markets. In such cases, neither the baseline nor the disaggregated coefficient necessarily admits a causal interpretation. Disaggregation invariance is nonetheless central: it ensures that introducing market-level controls does not mechanically change the estimand through the change in aggregation level. As a result, it becomes possible to interpret differences between baseline and disaggregated coefficients as reflecting the role of market shocks, rather than as artifacts of disaggregation. When disaggregation invariance fails, estimates obtained with market fixed effects may differ from firm-level estimates even if market shocks are irrelevant, making it impossible to assess whether disaggregation genuinely sharpens identification.

3.2 Why disaggregation invariance breaks

In the following, we show that disaggregation can alter the underlying estimand through three distinct channels: differences in implicit weights, nonlinear aggregation through Jensen's inequality, and the treatment of entry and exit.

Weights. Disaggregation changes how observations are weighted. Let $\mathcal{T}$ denote the set of treated firms, and let $m(i)$ denote the set of markets in which firm i operates. We assume away market entry and exit for now, so that firm-market growth rates are always defined. Let N_T be the number of treated firms and N_{Tm} the number of treated firm–market observations. At the firm level, the average growth rate of treated firms is

$$E(\Delta S_{it} \mid Treated = 1) = \frac{1}{N_T} \sum_{i \in \mathcal{T}} \Delta S_{it},$$

so that each treated firm receives equal weight. By contrast, at the firm–market level, ordinary least squares averages growth rates across treated firm–market observations:

$$E(\Delta S_{imt} \mid Treated = 1) = \frac{1}{N_{Tm}} \sum_{i \in \mathcal{T}} \sum_{m \in m(i)} \Delta S_{imt},$$

which assigns equal weight to each firm–market observation. Firms operating in more markets therefore receive greater weight.

For growth rate definitions under which firm-level growth is a weighted average of firm–market growth rates, this discrepancy can in principle be eliminated by reweighting firm–market observations. For the standard growth rate,

$$\Delta^s S_{it} = \sum_{m \in m(i)} w_{im}^s \, \Delta^s S_{imt}, \qquad w_{im}^s \equiv \frac{S_{im,t-h}}{S_{it-h}},$$

and for the midpoint growth rate,

$$\Delta^{mp} S_{it} = \sum_{m \in m(i)} w_{im}^{mp} \, \Delta^{mp} S_{imt}, \qquad w_{im}^{mp} \equiv \frac{S_{im,t} + S_{im,t-h}}{S_{it} + S_{it-h}}.$$

In both cases, firm-level growth is a linear aggregation of firm–market growth rates. Estimating equation (1) by weighted least squares using these weights restores equality between $E(\Delta S_{it} \mid Treated = 1)$ and $E(\Delta S_{imt} \mid Treated = 1)$.

Jensen's inequality. For log growth rates, weighting alone is not sufficient. Firm-level log growth satisfies

$$\Delta^l S_{it} = \log\left(\sum_{m \in m(i)} w_{im}^l \, \exp\Big(\Delta^l S_{imt}\Big) \right), \qquad w_{im}^l \equiv \frac{S_{im,t-h}}{S_{it-h}}. \tag{6}$$

Because the logarithm is concave, Jensen's inequality implies

$$\log\left(\sum_m w_{im}^l \exp(x_{imt}) \right) \; \geq \; \sum_m w_{im}^l x_{imt},$$

with strict inequality unless firm–market growth rates are identical across markets. As a result, disaggregation invariance fails for log growth rates even in WLS regressions.

Entry and exit. Entry and exit at the firm–market level introduce a third source of non-invariance. When firms enter new markets or exit existing ones, zeros arise in S_{imt} or $S_{im,t-h}$. As a result, log growth rates are defined only for markets in which the firm is active in both periods and therefore exclude entry and exit by construction. Standard growth rates are only defined in markets in which the firm was present at $t-h$. Firm-level growth, however, aggregates sales over all markets in which the firm operates in either period. Because firm-level and firm–market growth are computed over different domains, disaggregation invariance fails even under weighted least squares.

3.3 Restoring disaggregation invariance

This section discusses two approaches that restore disaggregation invariance, i.e., ensure that the disaggregated and baseline regressions identify the same estimand, $\beta_d = \beta_b$, in the absence of market-level shocks: midpoint growth rates and Poisson estimation.

Proposition 1 (Disaggregation invariance)**.** *Suppose firm-level sales decompose additively across markets, $S_{it} = \sum_m S_{imt}$. Then the disaggregated and baseline regressions identify the same estimand, $\beta_d = \beta_b$, under either specification: (i) midpoint growth rates estimated by weighted least squares with midpoint weights $w_{im}^{mp} = (S_{imt} + S_{im,t-h})/(S_{it} + S_{i,t-h})$; or (ii) the Poisson level specifications (7) and (8) with firm and firm–market fixed effects. Both accommodate firm–market entry and exit.*

Midpoint growth rates. The midpoint growth rate simultaneously addresses the three sources of non-invariance identified above: weighting, nonlinearity, and entry and exit. Two features of the midpoint growth rate are key. First, aggregation is linear: firm-level midpoint growth is a weighted average of firm–market midpoint growth rates. As a result, Jensen's inequality does not arise, in contrast to log growth rates. Second, the midpoint growth rate is well defined in the presence of zeros, ensuring that firm–market entry and exit are incorporated symmetrically. Firm-level growth therefore aggregates changes over the same domain as the firm–market data. We formally show that midpoint growth rates with midpoint weights satisfy disaggregation invariance in section D.2.

A natural concern is that midpoint weights depend on contemporaneous sales, which may themselves be affected by treatment. We discuss this issue in section C. We show that midpoint weights introduce no additional identification problem when treatment is randomly assigned at the firm level: in that case, the midpoint-weighted firm–market regression targets the same estimand as the firm-level regression. When identification instead relies on firm–market comparisons conditional on market fixed effects, midpoint weights can create bias only if they differentially weight idiosyncratic firm–market shocks for treated and control firms within the same market. We show that across the eight datasets used in section 2, midpoint weights are strongly correlated with standard weights, suggesting that firm–market innovations play a limited role in shaping midpoint weights.

In practice, a simple robustness check is to replace midpoint growth rates with standard firm–market growth rates and use predetermined weights w_{im}^{s}. As we explain in subsection 4.3, this specification identifies the effect of treatment on sales growth in markets in which the firm was already active. The contribution of treatment to market entry can then be recovered separately using a firm-level regression of entry rates on treatment. While this alternative does not incorporate intensive- and extensive-margin adjustments within a single regression, it provides a transparent robustness check based on predetermined weights while still allowing comparisons across aggregation levels.

Poisson estimation. Poisson estimation provides an alternative route to disaggregation invariance that does not rely on growth rates or on weights constructed from post-treatment outcomes. Instead, it models sales directly in levels and exploits the multiplicative structure of the conditional mean. Consider a two-period panel with one pre-treatment and one post-treatment observation. At the firm level, we estimate the Poisson regression

$$E(S_{it} \mid i, t) = \exp(\alpha_i + \mu_t + \beta_b \, Treated_i \times Post_t)\,, \tag{7}$$

where α_i is a firm fixed effect and μ_t a time fixed effect. At the firm–market level, we estimate

$$E(S_{imt} \mid i, m, t) = \exp(\alpha_{im} + \mu_t + \beta_d \, Treated_i \times Post_t)\,, \tag{8}$$

where α_{im} is a firm–market fixed effect.

Appendix D.1 shows that the first-order condition for β_d in the disaggregated regression is identical to the first-order condition for β_b in the baseline regression. As a result, the two specifications target the same estimand ($\beta_d = \beta_b$).

This equivalence follows from the multiplicative structure of the Poisson model. Because treatment shifts sales proportionally and by the same factor in all markets of a given firm, identification is inherently at the firm level. Firm–market fixed effects absorb all permanent differences in sales across markets, and the first-order condition aggregates naturally over markets. Replication across markets therefore affects precision but not the estimand, yielding disaggregation invariance by construction.

Inference. Our definition of disaggregation invariance involves an equality of the baseline and disaggregated estimand. What about standard errors? In subsection D.3, we show that, under the midpoint-growth specification, clustering standard errors at the firm level yields identical standard errors in the firm-level and firm–market-level regressions. Thus, our approach reconciles not only treatment effect estimates, but also inference across aggregation levels.

3.4 When does disaggregation invariance matter?

Disaggregation invariance is primarily a concern when the discrepancy between baseline and disaggregated coefficients is economically meaningful. In many applications, differences between firm-level and firm–market estimates may be small, in which case standard empirical practice remains a good approximation. In this section, we use simulations to quantify when disaggregation invariance is empirically important and to clarify the mechanisms behind the divergence between firm and firm–market estimates.

3.4.1 Data-generating process

We proceed with simulations to quantify the gap between firm-level and firm–market estimates under standard empirical specifications. Our data-generating process builds on the framework of Eaton, Kortum and Kramarz (2011) and delivers closed-form expressions for average treatment effects at various aggregation levels and for different growth rate definitions. Firms sell across multiple markets in two periods, and both market entry and sales conditional on entry depend on firm productivity. Treatment is modeled as a positive productivity shock, so that it affects sales through both the intensive margin (higher sales in existing markets) and the extensive margin (entry into additional markets and reduced exit). Treatment is randomly assigned at the firm level and is independent of initial productivity and all other firm characteristics.

While the model is motivated by export dynamics, its structure is much more general. Many economic environments feature outcomes that aggregate across multiple underlying units, so that treatment can affect both activity within a unit and the set of units over which activity takes place. Examples include banks allocating credit across borrowers, multi-establishment firms choosing where to operate across locations, or firms expanding employment across occupations.

Firm productivity. We consider a collection of firms indexed by i observed in two periods $t \in \{1, 2\}$. Each firm is characterized by productivity θ_{it}. In the pre-treatment period, initial productivity $\theta_{i,1}$ is heterogeneous and drawn uniformly from a finite set of values. To isolate the effect of treatment, we assume that half of firms are treated between periods 1 and 2. Treatment takes the form of a proportional productivity increase of size $g > 0$:

$$\theta_{i,2}^{Treated} = \theta_{i,1}(1 + g), \qquad \theta_{i,2}^{Control} = \theta_{i,1}.$$

Sales per market. Conditional on operating in a market, firm i draws sales in market m at time t from an exponential distribution:

$$E(S_{imt} \mid \theta_{it}) = \mu_0 \, \theta_{it}^{\gamma}.$$

The parameter γ governs the elasticity of sales to productivity and can be interpreted as capturing how strongly productivity translates into market-level revenue. When γ is small, sales are weakly related to productivity. One interpretation of a low γ could be the presence of economic frictions (e.g., switching costs) preventing firm competitive advantages from translating into revenue differences.

Number of markets (extensive margin). Firm productivity also affects the extensive margin. We model the number of active markets as[5]

$$E(K_{it} \mid \theta_{it}) = 1 + \lambda_0 \, \theta_{it}^{\eta}.$$

The parameter η governs how strongly market participation responds to productivity. Larger values of η imply that differences in productivity translate into larger differences in the number of markets in which firms operate, for instance due to the presence of important market fixed costs of entry.

Total firm sales. Firm-level sales aggregate across all markets in which the firm operates. Taking expectations yields

$$E(S_{it} \mid \theta_{it}) = E(K_{it} \mid \theta_{it}) \cdot E(S_{imt} \mid \theta_{it}) = (1 + \lambda_0 \, \theta_{it}^{\eta}) \, \mu_0 \, \theta_{it}^{\gamma}.$$

Thus, treatment increases firm-level sales through two channels: higher sales per market and expansion in the number of markets.

Implied average treatment effects. A key advantage of this simulation framework is that it generates sales in levels in both periods, so we can evaluate treatment effects under different growth rate transformations using the same underlying parameters. We compute the implied firm-level average treatment effect under log growth and midpoint growth, and we provide detailed derivations in Appendix B.[6]

These simulated firm-level average treatment effects provide natural benchmarks for our regression analysis. Because the data-generating process is fully specified, the implied ATE represents the true causal effect of treatment in the simulated economy. We can therefore assess the performance of standard empirical specifications by comparing regression estimates obtained at different aggregation levels to this benchmark.

Specification choice. For each source of disaggregation-invariance failure, namely implicit weights, Jensen's inequality, and entry and exit, we compare our midpoint-growth specification with midpoint weights to a standard OLS specification using log growth rates. We use OLS with log growth rates as our main benchmark because we view it as representative of common empirical practice. In Appendix E, we also compare the midpoint-growth specification with midpoint weights to the closest alternative specification for which disaggregation invariance fails. This second compari-

[5] We model the number of active markets as one plus a Poisson distribution to ensure that firms always operate in at least one market. This allows us to abstract from firm-level entry and exit, which would unnecessarily complicate the analysis.

[6] The average treatment effect is not defined for the standard growth rate under our simulation parameters.

Table 1: Disaggregation simulations: baseline parameters

Parameter	**Description**	**Default**
Initial firm productivity		
θ_1	Initial productivity	1
Firm–market sales: $S \sim \text{Exponential}(\mu_0\, \theta^\gamma)$		
μ_0	Baseline sales multiplier	1
γ	Sensitivity to productivity	1
Number of markets per firm: $K \sim \text{Poisson}(\lambda_0\, \theta^\eta) + 1$		
λ_0	Baseline market multiplier	1
η	Sensitivity to productivity	1
Treatment: $\theta_2 = (1+g) \cdot \theta_1$		
g_{control}	Productivity upgrade for control firms	0
g_{treated}	Productivity upgrade for treated firms	0.1
Simulation specifications		
N_{firm}	Number of firms per simulation	5,000
N_{loop}	Number of simulations	30

son isolates each source of failure of disaggregation invariance and makes it easier to assess the magnitude of the discrepancies across levels of aggregation when the other channels are shut down.

Simulation implementation. For each parameter configuration, we generate simulated datasets 30 times and each time estimate both the baseline firm-level regression and its disaggregated firm–market counterpart. We report the average estimated coefficient across repetitions for each specification, along with the corresponding standard errors computed from the dispersion of estimates across simulation runs. This procedure ensures that the reported gaps between baseline and disaggregated coefficients reflect systematic differences in estimands rather than simulation noise.

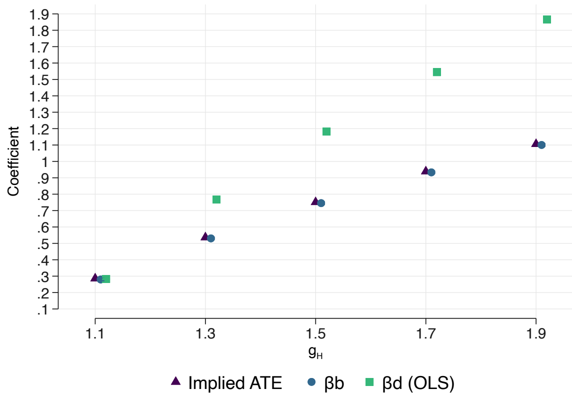
Unless stated otherwise, the disaggregation simulations use the parameters summarized in Table 1. Our parameter choices imply that on average in the pre-treatment period, firms are present in 2 markets and sell \$1 million per market.

3.4.2 Implicit weights

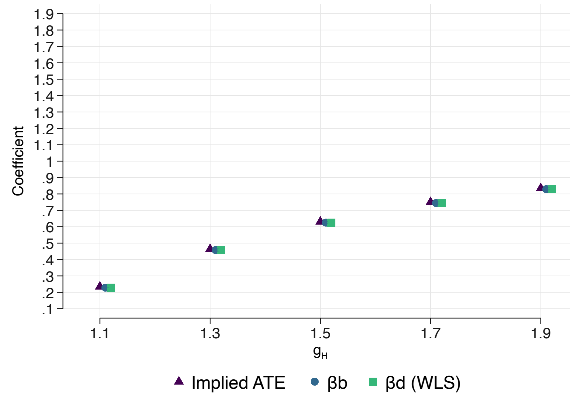
Differences in implicit weights need not matter if treatment effects are homogeneous across firms. When all firms experience the same causal effect, changing the unit of observation only redistributes weight across identical effects, so the estimand remains unchanged.

Implicit weights become important as soon as treatment effects become more heterogeneous. To illustrate, consider a simple simulation in which firms draw initial productivity from two values, $\theta \in \{\theta_L, \theta_H\}$ with $\theta_H > \theta_L$. Our data-generating process implies that more productive firms operate in more markets on average. To abstract from the extensive margin, we let the number of

Figure 3: Disaggregation invariance: Implicit weights



(a) Log growth rate – OLS



(b) Midpoint growth rate – WLS

Note. The simulation follows Table 1. Firms draw low or high initial productivity with equal probability. Low-productivity firms have $\theta_L = 0.1$, and high-productivity firms have $\theta_H = 3$. Pre-period productivity determines the number of markets, with $\eta = 3$. The number of markets is then held fixed across periods. Treated low-productivity firms receive productivity multiplier $g_L = 1.1$. Treated high-productivity firms receive the multiplier g_H shown on the x-axis. Control firms receive no productivity change.

markets be determined by pre-period productivity and remain fixed across periods. In the second period, treated firms experience a productivity upgrade that depends on their initial productivity, with $g_H > g_L$, while control firms experience no productivity change. Sales are realized at the firm–market level and aggregated to the firm level.

Because more productive firms operate in more markets, disaggregated regressions mechanically place greater weight on high-productivity firms. The firm–market coefficient therefore over-represents the effect for firms with $\theta = \theta_H$, whereas the firm-level coefficient assigns equal weight to each firm. More generally, when treatment effects covary with firms' number of markets, the two aggregation levels identify different weighted averages of heterogeneous effects, with the disaggregated estimate tilted toward firms that generate more firm–market observations.

Figure 3 illustrates this mechanism by holding $g_L = 1.1$ and letting g_H vary. We use $\eta = 3$ to make the number of markets strongly dependent on productivity. Panel (a) uses log growth rates estimated by OLS, while panel (b) uses midpoint growth rates estimated by WLS.[7] In panel (a), the firm–market coefficient from unweighted log-growth regressions increasingly diverges from the firm-level coefficient as treatment heterogeneity rises, reflecting the greater weight placed on high-productivity firms. The firm- and firm-market coefficients only coincide when treatment is homogeneous, that is, $g_H = g_L$. In contrast, panel (b) shows that midpoint growth rates estimated with midpoint weights yield identical coefficients across aggregation levels, closely matching the implied firm-level average treatment effect.

[7] Figure E.1 compares OLS and WLS estimates using midpoint growth rates and shows that WLS is necessary to achieve disaggregation invariance.

3.4.3 Jensen inequality

For log growth rates, firm-level growth is not a linear aggregation of firm–market growth rates. Rewriting equation (6), the firm-level growth rate can be expressed as the sum of the weighted average of firm–market growth rates and an additional concavity term:

$$\Delta^l S_{it} = \sum_{m \in m(i)} w^l_{im} \, \Delta^l S_{imt} \; + \; C_{it},$$

where the concavity term is defined as

$$C_{it} \equiv \log\left(\sum_{m \in m(i)} w^l_{im} \exp(\Delta^l S_{imt}) \right) - \sum_{m \in m(i)} w^l_{im} \Delta^l S_{imt}.$$

The concavity term can be better understood by performing a second-order approximation around the weighted mean $\bar{\Delta}_{it} \equiv \sum_m w^l_{im} \Delta^l S_{imt}$:

$$C_{it} \approx \frac{1}{2} \sum_{m \in m(i)} w^l_{im} \left(\Delta^l S_{imt} - \bar{\Delta}_{it} \right)^2 = \frac{1}{2} \operatorname{Var}_w \left(\Delta^l S_{imt} \right), \tag{9}$$

where $\operatorname{Var}_w(\cdot)$ denotes the within-firm variance of firm–market log growth rates under weights w^l_{im}. Thus, firm-level log growth differs from the weighted average of firm–market log growth because it incorporates not only the mean of market-level growth, but also its dispersion across markets.

This approximation clarifies the condition required for disaggregation invariance under log growth. Even if firm–market observations are correctly weighted, equation (5) implies that firm-level and firm–market coefficients coincide (abstracting from higher-order terms) if

$$E\Big[\operatorname{Var}_w(\Delta^l S_{imt}) \mid Treated = 1 \Big] = E\Big[\operatorname{Var}_w(\Delta^l S_{imt}) \mid Treated = 0 \Big].$$

Disaggregation invariance, in other words, fails whenever treated and control firms exhibit different within-firm variances of firm–market growth rates (i.e., treatment-induced heteroskedasticity). For instance, if treatment increases the dispersion of firm–market growth rates across markets, then C_{it} rises for treated firms and log-growth regressions identify different estimands across aggregation levels.

To isolate this mechanism, we consider a simple simulation in which all firms operate in the same number of markets in both periods, so there is no extensive-margin adjustment and no role for implicit weights. We set the number of markets to five to ensure sufficient within-firm variations in sales growth. Firm–market sales are independent draws from a Gamma distribution,

$$S_{imt} \sim \Gamma(k_t, \lambda_t),$$

where k_t is the shape parameter and λ_t the rate parameter. Control firms have (k_1, λ_1) in both periods, while treated firms have (k_1, λ_1) in period 1 and (k_2, λ_1) in period 2. Relative to the exponential benchmark, the Gamma specification allows the variance of log growth across firm–market observations to differ between treated and control firms and therefore to shift the concavity term.[8]

Figure 4 illustrates the mechanism by varying k_2 while holding $k_1 = \lambda_1 = 1$.[9] Panel (a) uses log growth rates estimated by WLS, while panel (b) uses midpoint growth rates estimated by WLS. Increasing k_2 raises expected sales per market, leading to larger average treatment effects, but it also reduces the within-firm variance of firm–market log growth for treated firms, making the concavity term more negative. Accordingly, in panel (a), both the firm-level and firm–market coefficients from log-growth regressions increase with k_2, capturing larger treatment effects. At the same time, the firm-level coefficient becomes progressively smaller than the firm–market coefficient, as the concavity term becomes more negative. The two coefficients coincide only when $k_1 = k_2$, that is, when treated and control firms have the same within-firm variance of firm–market growth rates. In panel (b), midpoint growth rates continue to deliver identical coefficients at the firm and firm–market levels.

3.4.4 Entry and exit

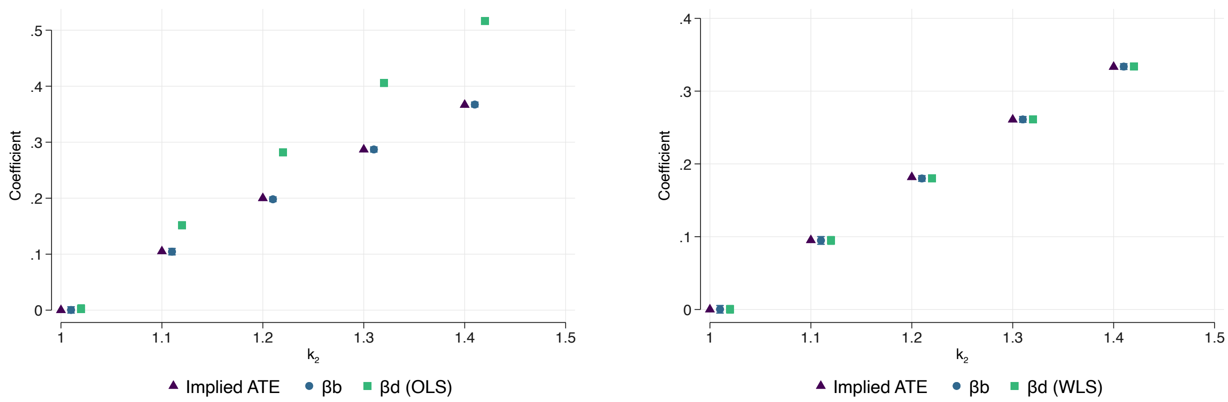
With log growth rates, firm–market regressions are defined only for markets in which the firm is continuously present in both periods. As a result, disaggregated log-growth specifications mechanically exclude entry and exit adjustments. To clarify when this affects disaggregation invariance, it is useful to decompose firm-level sales growth into an intensive-margin component, reflecting changes in sales in continuing markets, and an extensive-margin component, reflecting entry into new markets and exit from existing ones. In particular, the midpoint growth rate satisfies

$$2\,\frac{S_{it} - S_{i,t-h}}{S_{it} + S_{i,t-h}} = \underbrace{2\,\frac{E_{it}}{S_{it} + S_{i,t-h}}}_{\text{Entry}} + \underbrace{2\,\frac{-X_{it}}{S_{it} + S_{i,t-h}}}_{\text{Exit}} + \underbrace{2\,\frac{C_{it} - C_{i,t-h}}{S_{it} + S_{i,t-h}}}_{\text{Intensive}}. \tag{10}$$

[8] One can show that the volatility of log growth under the Exponential baseline is equal to $\pi^2/3$, which is therefore the same for treated and control firms.

[9] Figure E.2 compares log and midpoint growth rates estimates using WLS and shows that the midpoint growth rate is necessary to achieve disaggregation invariance.

Figure 4: Disaggregation invariance: Jensen's inequality



(a) Log growth rate – OLS

(b) Midpoint growth rate – WLS

Note. The simulation follows Table 1. All firms operate in five markets in both periods. Firm–market sales are independent Gamma draws with shape k_t and rate λ_t. Control firms keep the same distribution in both periods. Treated firms start from the same distribution as controls, but in period 2 their sales are drawn with shape parameter k_2, shown on the x-axis. The remaining Gamma parameters are fixed at $k_1 = \lambda_1 = 1$.

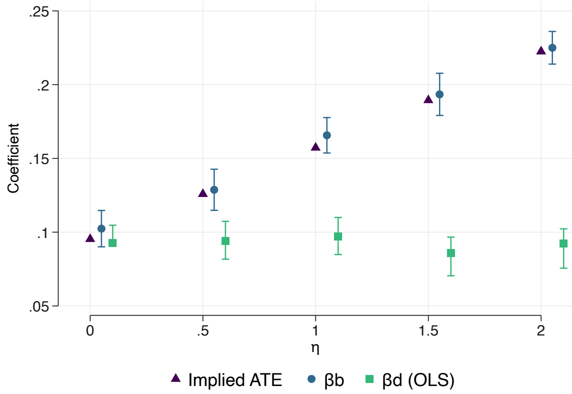
where E_{it} and X_{it} denote sales associated with entry and exit, respectively, and C_{it} denotes sales in continuing markets. Equation (5) therefore implies that disaggregation invariance holds as long as treatment affects only the intensive margin and does not systematically shift the entry or exit margins.

In our simulation framework, the sensitivity of the extensive margin to treatment is governed by the parameter η in the market-entry process described in Section 3.4.1. When $\eta = 0$, firms still operate in multiple markets, but market participation is independent of productivity and therefore unaffected by the treatment-induced productivity shock. As a result, the extensive margin is present but does not contribute to the treatment effect. As η increases, treatment induces greater expansion into new markets and reduces exit, so the extensive margin becomes increasingly important in shaping the average treatment effect. To isolate the role of the extensive margin, we use weights for both the log and midpoint growth rate regressions and impose homogeneous treatment effects, shutting down the implicit-weights and Jensen channels.

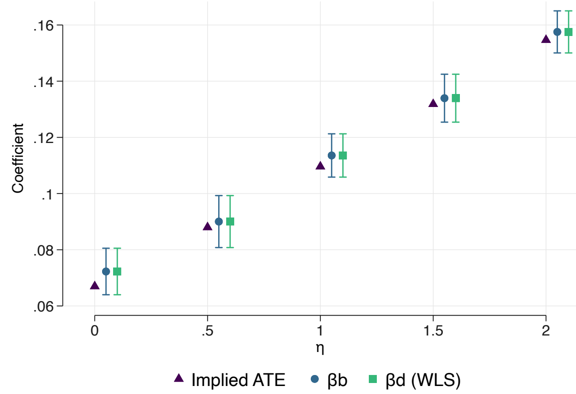
Figure 5 shows the estimation results for different values of η.[10] Panel (a) uses log growth rates estimated by WLS, while panel (b) uses midpoint growth rates estimated by WLS. When $\eta = 0$, the two coefficients are essentially identical, consistent with treatment operating purely through the intensive margin. As η increases, the gap between firm-level and firm–market coefficients widens monotonically, as disaggregated specifications increasingly miss part of the entry and exit response. Panel (b) shows that the midpoint growth rate always delivers identical firm- and firm–market estimates, which closely match the implied average treatment effect.

[10] Figure E.3 compares log and midpoint growth rates estimates using WLS and shows that the midpoint growth rate is necessary to achieve disaggregation invariance.

Figure 5: Disaggregation invariance: entry and exit



(a) Log growth rate – OLS



(b) Midpoint growth rate – WLS

Note. The simulation follows Table 1. The parameter η, which governs how strongly market participation responds to productivity, varies on the x-axis.

When is the extensive margin likely to threaten disaggregation invariance? A simple way to assess the role of the extensive margin in the average treatment effect is to run separate firm-level regressions of the different components of equation (10) on the treatment indicator. Table 2 shows that this methodology delivers an exact decomposition of the average treatment effect into entry, exit, and intensive components, a property of midpoint growth rates already emphasized in Beaumont, Libert and Hurlin (2019); Fonseca and Matray (2024); Matray, Müller, Xu and Kabir (2024). Importantly, the results show that the extensive component (entry plus exit) increases with η, confirming that larger values of η increase the probability of entering new markets and reduces the probability of leaving existing ones.

Table 2: Decomposition into intensive and extensive margins

η	ATE	Total	Intensive margin	Entry	Exit
0.00	0.07	0.07	0.06	0.00	0.01
		[0.03]	[0.02]	[0.01]	[0.01]
0.50	0.09	0.09	0.06	0.02	0.02
		[0.03]	[0.02]	[0.01]	[0.01]
1.00	0.11	0.11	0.06	0.03	0.02
		[0.03]	[0.03]	[0.01]	[0.01]
1.50	0.13	0.13	0.05	0.05	0.03
		[0.03]	[0.02]	[0.01]	[0.01]
2.00	0.15	0.16	0.06	0.06	0.04
		[0.03]	[0.02]	[0.01]	[0.01]

Note. The simulation follows Table 1. The parameter η, which governs the response of market participation to productivity, varies across rows. The table reports the implied average treatment effect and firm-level regressions of total midpoint growth, the intensive-margin component, the entry component, and the exit component on treatment.

3.5 Application: detection of confounding shocks

The previous simulations show that when disaggregation invariance fails, disaggregated specifications can yield different estimates even in the absence of confounding shocks. In turn, when disaggregation invariance holds, comparisons across levels of aggregation become informative about confounding shocks and their contribution to the baseline average treatment effect. This subsection illustrates how researchers can leverage disaggregation invariance for this purpose.

We introduce confounding shocks by introducing market-level shocks that are correlated with treatment status. Specifically, each market j is subject to a multiplicative shock in the second period, so that firm–market sales satisfy

$$S_{ij,2} = X_{ij,2} \times \text{Shock}_j,$$

where $\log(\text{Shock}_j)$ is normally distributed. In this setting, the log-growth specification admits an additive decomposition, so that market fixed effects can absorb the confounding shock.

Treatment is assigned at the firm level as a function of firms' average pre-period exposure to these market shocks, so that treated firms are more likely to operate in markets with systematically different realizations of Shock_j. However, conditional on market fixed effects, treatment remains as good as randomly assigned across firms, since treatment depends only on market-level shocks, which are fully absorbed by the fixed effects, and not on firm–market residual shocks.

For illustrative purposes, we consider a setting with a relatively high sensitivity of market participation to productivity ($\eta = 3$) and a negative correlation between treatment and market shocks, so that treated firms are more likely to operate in markets experiencing negative shocks.

A common empirical approach is to compare firm-level estimates with firm–market estimates that include market fixed effects, with larger gaps suggesting more substantial confounding shocks. We compare the ability of standard practice, namely log growth rates estimated by OLS in panel (a), and our approach, namely midpoint growth rates estimated by WLS in panel (b), to detect these confounding shocks.

Table 3 shows that, under the log-growth specification, firm-level and firm–market estimates are very similar even in the presence of confounding shocks. In the simulation, both coefficients are far from the true average treatment effect, yet their similarity may lead the researcher to conclude that market-level shocks are not a concern. This result arises because log-growth specifications fail to capture extensive-margin adjustments. As shown in the previous subsection, disaggregated log regressions omit observations corresponding to entry and exit, leading to a downward bias in the estimated treatment effect. When market shocks are negatively correlated with treatment, firm-level estimates are also biased downward. The two biases are of similar magnitudes, causing

Table 3: Disaggregation invariance: Detecting confounding shocks

Specification	Mean	SE
Panel A: Log growth rate – OLS		
ATE	0.3264	–
$\hat{\beta}^b$	0.1140	0.1720
$\hat{\beta}^d$, no FE	-0.2245	0.2442
$\hat{\beta}^d$, market FE	0.1003	0.0466
Panel B: Midpoint growth rate – WLS		
ATE	0.2267	–
$\hat{\beta}^b$	0.0744	0.1085
$\hat{\beta}^d$, no FE	0.0744	0.1085
$\hat{\beta}^d$, market FE	0.2276	0.0439

Note. The simulation follows Table 1 and adds market shocks in period 2. Market shocks are multiplicative: $S_{ij,2} = X_{ij,2} \times Shock_j$, with $\log(Shock_j) \sim N(0,1)$. Treatment is assigned at the firm level using firms' average pre-period exposure to market shocks. The correlation parameter is -0.3, so treated firms are more exposed to negative shocks. The table reports the implied ATE and the mean and standard errors of estimated coefficients across simulations. Firm–market regressions use OLS with log growth rates and WLS with midpoint growth rates. "Market FE" adds market fixed effects.

firm-level and firm–market coefficients to appear nearly identical.

By contrast, midpoint growth rates preserve disaggregation invariance and therefore provide a reliable diagnostic for the presence of confounding shocks. The firm–market coefficient increases sharply when market fixed effects are included, correctly indicating that market shocks are negatively correlated with treatment and bringing the estimate close to the implied average treatment effect. An additional advantage of disaggregation invariance is that, because firm-level and firm–market coefficients are aligned, the magnitude of the bias induced by confounding shocks can be directly inferred. In this case, failing to account for market shocks leads to an underestimation of the coefficient by 69%.

4 Aggregation

4.1 Aggregation invariance

We now formalize the notion of aggregation invariance. Let $j(i)$ denote firm i's industry. Firm-level sales S_{it} aggregate into industry-level sales

$$S_{jt} = \sum_{i:\, j(i)=j} S_{it},$$

and we define industry growth ΔS_{jt} using the same growth rate as at the firm level (log, standard, or midpoint). Because industries combine firms with potentially different treatment statuses, moving from the firm to the industry level requires defining industry exposure to treatment:

$$Share\,Treated_j = \sum_{i:\, j(i)=j} \omega_{ij}\, Treated_i.$$

For the firm-level regression to aggregate properly into the industry-level regression, these weights must coincide with those governing the aggregation of growth rates on the left-hand side. Moreover, since industries are typically given equal importance in the aggregated regression, the weights must add up to one at the industry level. When growth is a linear aggregation of lower-level outcomes, this requirement directly pins down the appropriate weights. In particular, for the standard and midpoint growth rates, the weights take the form

$$\omega_{ij}^{s} = \frac{S_{i,t-h}}{S_{j,t-h}}, \quad \omega_{ij}^{mp} = \frac{S_{it} + S_{i,t-h}}{S_{jt} + S_{j,t-h}}.$$

By convention, we also use the weights associated with the standard growth rate when using the log growth rate.

Aggregation replaces the firm-level outcome $\{\Delta S_{it}\}_i$ by the industry-level outcome $\{\Delta S_{jt}\}_j$ and estimates the specification in Equation 3. We take the firm as the baseline unit and the industry as the aggregated unit, but the same logic applies whenever a baseline outcome aggregates exactly across units.

Exogeneity assumptions. To interpret differences between firm- and industry-level coefficients as evidence of spillovers, exogeneity must hold at both levels of aggregation. In particular, the firm-level moment condition

$$E[e_{it}\, Treated_i] = 0 \tag{11}$$

must hold, as well as its industry-level counterpart

$$E[e_{jt}\, Share\,Treated_j] = 0. \tag{12}$$

Importantly, exogeneity at the firm level does not in general imply exogeneity at the industry level. Even when the firm-level moment condition holds, the distribution of treated firms across industries may be correlated with industry-level shocks or treatment-effect heterogeneity, leading to biased industry-level estimates.[11]

[11] For instance, treatment may satisfy the firm-level moment condition even though treatment effects are heterogeneous. Suppose financially constrained firms react less to treatment. If industries with a higher share of treated firms also have a higher share of financially constrained firms, then the industry-level regressor $Share\,Treated_j$ will be correlated with industry-level treatment-effect heterogeneity. The industry-level coefficient will then differ from the average firm-level effect even though the firm-level moment condition holds.

SUTVA. Disaggregation decomposes a firm-level outcome into within-firm components, so all disaggregated observations share the same treatment status. Aggregation, by contrast, sums outcomes across firms with potentially different treatment statuses. Industry growth therefore reflects not only the direct effect on treated firms, but also interactions between treated and control firms within the same industry.

The fact that aggregation captures such interactions is precisely why researchers aggregate in the first place: comparing firm- and industry-level estimates is a standard way to detect business stealing, reallocation, or other spillovers within industries. For this comparison to be meaningful, however, the empirical specification must ensure that the firm- and industry-level estimands coincide in the absence of SUTVA violations. Otherwise, differences across aggregation levels may arise mechanically from the change in specification rather than from cross-firm interactions.

These considerations motivate our definition of aggregation invariance. Let β^b denote the estimand from the firm-level regression and β^a the estimand from the industry-level regression. We say that *aggregation invariance* holds if $\beta^a = \beta^b$ in the absence of SUTVA violations.[12]

4.2 Why aggregation invariance breaks

A key difference from disaggregation is that aggregation changes the regressor, not only the level of observation. As a result, the firm- and industry-level regressions do not rely on the same population equation.

To see this, note that under disaggregation invariance, the population moment condition underlying the firm-level regression can be written as the sum of corresponding disaggregated moments:

$$E[e_{it}\,Treated_i] = E\left[\sum_{m\in m(i)} \omega_{imt} e_{imt}\,Treated_i\right] = 0.$$

The baseline and disaggregated regressions therefore rely on the same underlying population equation, so disaggregation invariance can be achieved by appropriately specifying the aggregation operator, as discussed in subsection 3.3.

Aggregation is different because it involves two distinct regressors: $Treated_i$ at the firm level and $Share\,Treated_j$ at the industry level. As a result, it relies on two distinct population moment conditions, one at the firm level (Equation 11) and one at the industry level (Equation 12). Importantly, the firm-level condition does not imply its industry-level counterpart, since the latter embeds interactions across firms within industries. Absent additional restrictions, the correspond-

[12] In principle, aggregation invariance can be defined without imposing exogeneity. For instance, it can be used in purely accounting exercises to relate firm-level covariances to aggregate outcomes. In this paper, however, we explicitly use aggregation invariance to detect violations of SUTVA, which requires a causal interpretation of the coefficients and therefore motivates imposing exogeneity at both levels.

ing estimands need not coincide, even in the absence of spillovers.

Aggregation invariance will also fail for the same mechanical reasons as disaggregation invariance. First, differences in implicit weights arise because industries differ in the number and size of firms, so aggregation reweights observations relative to the firm-level regression. Second, when the growth operator is nonlinear, as with log growth, Jensen's inequality implies that industry growth is not a linear aggregation of firm-level growth rates. Third, when treatment affects entry and exit, industry-level outcomes reflect changes in the set of firms, so aggregation may capture extensive-margin adjustments that are not aligned with the firm-level specification. For brevity, we do not revisit these mechanisms in detail. Instead, we assess their quantitative importance in the aggregation context using simulations in subsection 4.4. Before turning to these simulations, however, we study in the next subsection how aggregation invariance can be restored.

4.3 Restoring aggregation invariance

In this subsection, we explain why the methods used to achieve disaggregation invariance (Poisson and midpoint growth rates) cannot restore aggregation invariance. We then present our approach to (i) align the baseline and aggregated population moment equations and (ii) accommodate entry and exit.

Poisson. Under the firm-level Poisson specification,

$$E(S_{it} \mid i, t) = \exp(\alpha_i + \mu_t + \beta_b \, Treated_i \times Post_t) \, .$$

To see why Poisson does not deliver aggregation invariance, consider an industry j with two firms, firm 0 (untreated) and firm 1 (treated). Aggregating across firms yields

$$E(S_{jt} \mid j, t) = \exp(\mu_t) \left[\exp(\alpha_0) + \exp(\alpha_1 + \beta_b \, Post_t)\right] .$$

The post-period treatment component is therefore governed not by the unweighted share of treated firms, but by the treated firm's share of predicted baseline sales:

$$\frac{\exp(\alpha_1)}{\exp(\alpha_0) + \exp(\alpha_1)} .$$

This object depends on the firm fixed effects rather than only on treatment status. As a result, the industry-level conditional mean cannot, in general, be written as a function of $Share\,Treated_j$ alone.

Midpoint growth rate. When using the midpoint growth rate, restoring aggregation invariance would require defining industry treatment exposure using midpoint weights, so that $Share\,Treated_j$ reflects the same weights as those governing the aggregation of outcomes. However, these weights depend on contemporaneous sales, $S_{it} + S_{i,t-h}$, and are therefore affected by treatment. As a result, the regressor $Share\,Treated_j$ becomes a function of the outcome, leading to a form of simultaneity bias.

This issue does not arise in the disaggregation setting. When disaggregating, midpoint weights are used only to aggregate firm–market outcomes back to the firm level, and the treatment variable remains fixed at the firm level. In particular, treatment is common across all firm–market observations for a given firm, so weighting does not alter the variation used for identification.

Aligning the population equations. For the sake of simplicity, we temporarily assume zero entry, so that aggregation of firm-level growth rates can be achieved using the standard growth rate. To align the firm- and industry-level population equations, we augment the firm-level regression by including industry treatment exposure:

$$\Delta^s S_{it} = \alpha + \beta^b\, Treated_i + \delta\, Share\,Treated_j + \varepsilon_{it}.$$

The associated population moment conditions are

$$E[\varepsilon_{it}\, Treated_i] = 0, \quad E[\varepsilon_{it}\, Share\,Treated_j] = 0.$$

Using the fact that $Share\,Treated_j$ is constant within industries and that outcomes aggregate linearly, the second condition implies

$$E[e_{jt}\, Share\,Treated_j] = 0,$$

so that the augmented firm-level regression and the industry-level regression rely on the same population equation. As we show in subsection D.4, this implies that the corresponding coefficients satisfy

$$\beta^a = \beta^b + \delta.$$

Thus, controlling for $Share\,Treated_j$ at the firm level aligns the population equations and ensures that the firm- and industry-level regressions target the same estimand.[13]

[13] An alternative approach would be to include industry fixed effects in the firm-level regression. In that case, the coefficient on $Treated_i$ would be identified solely from within-industry variation and would correspond to β^b. The industry-level coefficient would remain equal to $\beta^b + \delta$, so that comparing firm- and industry-level coefficients would recover δ.

This specification also has a natural economic interpretation. It decomposes treatment effects into a direct effect, captured by $Treated_i$, and an indirect effect operating through exposure to treated firms within the same industry, captured by $ShareTreated_j$. When $\delta = 0$, firm outcomes do not depend on the treatment status of other firms. In this case, the augmented firm-level regression reduces to the baseline specification and we obtain $\beta^a = \beta^b$. This corresponds exactly to aggregation invariance: firm- and industry-level regressions yield the same estimand in the absence of SUTVA violations.

Including the share of treated firms at the firm level is often referred to as the "exposure-to-treated" approach and is commonly used as an alternative to aggregation to detect SUTVA violations (e.g., Crépon, Duflo, Gurgand, Rathelot and Zamora, 2013; Cai and Szeidl, 2024; Lerche, 2025). The decomposition $\beta^a = \beta^b + \delta$ provides a direct link between the aggregation approach and the exposure-based approach. In particular, it shows that comparing firm- and industry-level estimates is equivalent to testing whether exposure to treated firms affects outcomes, that is, whether $\delta = 0$. Accordingly, the two approaches coincide when aggregation invariance holds.

Accommodating the extensive margin. We favor the midpoint growth rate for disaggregation because it naturally accommodates entry and exit. Since the midpoint growth rate cannot be used for aggregation, we instead rely on an explicit decomposition to account for the extensive margin. The starting point is the accounting identity

$$\Delta^s S_{jt} = \frac{Sales_{jt}^{entry}}{S_{j,t-h}} + \frac{Sales_{jt}^{inc} - Sales_{j,t-h}^{inc}}{S_{j,t-h}},$$

where $Sales_{jt}^{entry}$ denotes total sales at time t by firms that are not present in $t-h$, and $Sales_{jt}^{inc}$ denotes total sales by incumbent firms, that is, firms with nonzero sales at $t-h$.

We first recover the intensive-margin component using a firm-level regression restricted to incumbents. Let $s_{ij,t-h} \equiv S_{i,t-h}/S_{j,t-h}$ denote firm i's lagged share in industry j. We estimate the augmented specification for incumbent firms (i.e., with nonzero lagged sales)

$$\Delta^s S_{it} = \alpha + \beta^b \, Treated_i + \delta \, ShareTreated_j + \varepsilon_{it},$$

by weighted least squares with weights $s_{ij,t-h}$. This weighting scheme mirrors the aggregation on the left-hand side and excludes entrants by construction, as they have zero sales at $t-h$. The coefficient β^b captures the direct effect of treatment on incumbents, while δ captures indirect effects operating through exposure to treated firms within the same industry.

We then recover the extensive-margin component using an industry-level regression. Let $Entry_{jt} \equiv$

$Sales^{entry}_{jt}/S_{j,t-h}$ denote the contribution of entry to industry growth. We estimate

$$Entry_{jt} = \alpha + \beta^E \, Share\,Treated_j + u_{jt},$$

which captures how industry-level exposure to treatment affects entry.

Combining the two components yields the industry-level treatment effect. Using the result from subsection D.4, the intensive-margin contribution at the industry level is given by $\beta^b + \delta$, so that the overall industry-level coefficient satisfies

$$\beta^a = \beta^b + \delta + \beta^E.$$

Thus, the aggregate effect decomposes into the direct effect on incumbents, indirect effects through within-industry exposure, and the contribution of entry. This two-step procedure aligns the sources of variation across aggregation levels while preserving a transparent decomposition of the aggregate treatment effect.

Proposition 2 (Aggregation invariance)**.** *Abstracting from entry, augmenting the firm-level regression with the industry treatment share $Share\,Treated_j$ and estimating by weighted least squares with weights $\omega^s_{ij} = S_{i,t-h}/S_{j,t-h}$ yields $\beta^a = \beta^b + \delta$. Absent SUTVA violations, $\delta = 0$ and $\beta^a = \beta^b$. With entry, the two-step decomposition gives $\beta^a = \beta^b + \delta + \beta^E$, where β^E is the coefficient from the industry-level entry regression.*

Inference. In subsection D.5, we show that the standard errors for $\beta_b + \delta + \beta^E$ and β_a are identical when regressions are clustered at the industry level. As in the disaggregation case, our approach therefore preserves inference across aggregation levels.

4.4 When does aggregation invariance matter?

This section studies when aggregation invariance is likely to matter quantitatively in practice. We use simulations to evaluate the different channels through which firm- and industry-level estimates can diverge under standard empirical specifications. We first study the role of implicit weights and then turn to Jensen's inequality. We do not run separate simulations for the entry margin, as the underlying intuitions are identical to those discussed in subsubsection 3.4.4. Instead, we illustrate the role of entry in an aggregation setting in subsection 4.5.

4.4.1 Data-generating process

We proceed with simulations to quantify the gap between firm-level and industry-level estimates under standard empirical specifications. Our data-generating process mirrors the one described in subsubsection 3.4.1, with three key differences. First, firms no longer operate across multiple markets. The exponential process used to model firm–market sales is therefore applied directly to total firm sales. Second, firms are assigned to industries, which differ in their exposure to treatment through the share of treated firms, drawn from a uniform distribution. Third, at time 2, new firms enter industries with probability p_{entry}. We set the number of potential entrants to 50. Entrants are assigned the same productivity as incumbent firms at time 1. For simplicity, we do not model firm exit. Table 4 reports the baseline parameters used in the aggregation simulations.

Table 4: Aggregation simulation: baseline parameters

Parameter	**Description**	**Default**
Initial firm heterogeneity		
θ_1	Initial productivity	1
Firm sales: $S \sim \text{Exponential}(\mu_0\, \theta^{\gamma})$		
μ_0	Baseline sales multiplier	1
γ	Sensitivity to productivity	1
Treatment: $\theta_2 = (1+g) \cdot \theta_1$		
g_{control}	Productivity upgrade for control firms	0
g_{treated}	Productivity upgrade for treated firms	0.1
Industry-level share of treated firms		
Share Treated$_j$	Probability that firm i within industry j is treated	U(0.1, 0.9)
Entry $p_{\text{entry}} = \kappa \cdot \textit{Share Treated}_j$		
N_e	Number of potential entrants	50
κ	Sensitivity of entry probability to industry treatment share	0.4
θ_e	Productivity of new entrants	1
Simulation specifications		
N_{ind}	Number of industries per simulation	100
$N_{\text{firm/ind}}$	Number of firms per industry	100
N_{sim}	Number of simulations	50

As in the case of disaggregation, we compare our midpoint-growth specification with midpoint weights to a standard OLS specification using log growth rates. In the aggregation simulations, we include the share of treated firms in the firm-level regression even under log growth rates. This allows us to show that baseline and industry-level estimates can differ even after the population equations have been aligned. The exception is the empirical exercise in subsection 4.5, where we omit the share of treated firms from the firm-level regression because the objective is to compare our approach to common empirical practice.

We also do not report implied average treatment effects in this section. The industry-level average treatment effect can be computed in a manner similar to subsection 3.1. However, aggregation is typically used not to estimate an industry-level ATE per se, but to compare firm- and industry-level estimates as a way to detect potential SUTVA violations. For this reason, we view the comparison between aggregation levels, rather than the industry-level ATE, as the relevant object.

4.4.2 Implicit weights

We now show how aggregation can break invariance through differences in implicit weighting across levels of aggregation. Throughout, we compare the industry-level coefficient β^a, obtained from the regression of industry growth on $ShareTreated_j$, to the firm-level counterpart $\beta^b + \delta$, obtained from the augmented firm-level specification in subsection 4.3. There are two distinct reasons why implicit weights may lead β^a and $\beta^b + \delta$ to diverge.

First, when using OLS, more populated industries receive more weight in firm-level regressions. As a result, firm-level estimates may differ from industry-level ones if industries with more firms respond differently to treatment. This mechanism is identical to the one discussed in subsubsection 3.4.2, so we do not elaborate further here.

Second, aggregation introduces an additional source of weighting that is specific to the industry level. At the firm level, an unweighted regression assigns equal importance to each firm. By contrast, industry-level outcomes aggregate firm sales, so firms with larger sales contribute more to industry growth. Industry regressions therefore implicitly weight firms by their size.

This implicit weighting creates a wedge between firm- and industry-level estimates whenever large firms behave differently, either because of heterogeneous treatment effects or because they are subject to idiosyncratic shocks. In concentrated industries, where a small number of firms account for a large share of industry activity, this mechanism can be quantitatively important, as industry-level estimates primarily reflect the behavior of these dominant firms rather than that of the average firm.

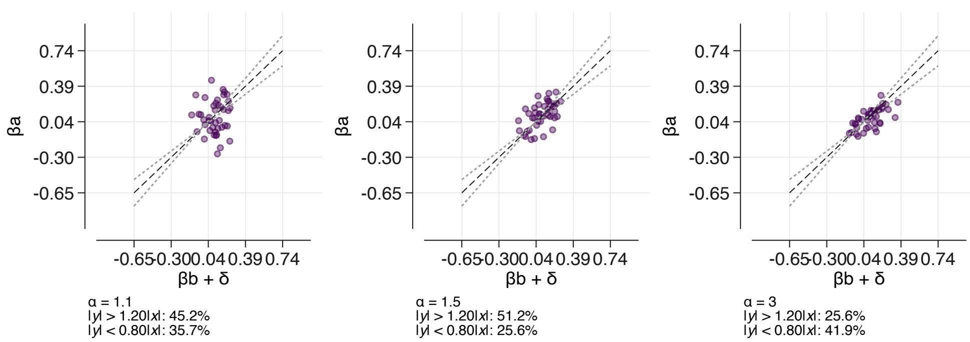
We illustrate this mechanism using the following simulation. We focus on the intensive margin by assuming that all firms are present at both periods. We maintain the assumption of homogeneous treatment and instead focus on the role of idiosyncratic shocks. Because sales are drawn from an exponential distribution, realized growth rates already contain idiosyncratic variation across firms. To capture differences in industry concentration, we draw initial firm productivity from a Pareto distribution,

$$\theta_{i,t_1} \sim \text{Pareto}(\alpha, \theta_{\min}),$$

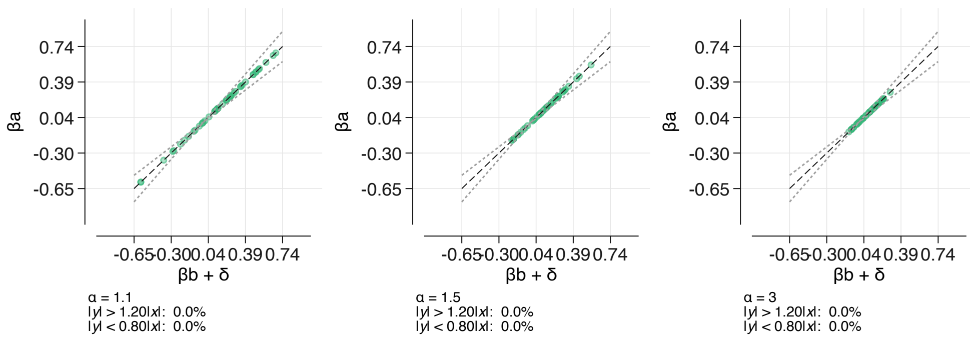
where the tail index α governs concentration. Smaller values of α generate a heavier right tail, leading to highly concentrated industries in which a few firms dominate sales. As α increases, the distribution becomes less skewed and firms are more similar in size. We rescale the lower bound

$\theta_{\min}$ so that average productivity is held fixed across specifications.

Figure 6: Aggregation invariance: Implicit weights



(a) Log growth rate – OLS



(b) Standard growth rate – WLS

Note. The figure plots $\beta^b + \delta$ on the x-axis and β^a on the y-axis across simulations for different values of α. The simulation follows Table 4. Initial firm productivity follows a Pareto distribution, $\theta_{i1} \sim \text{Pareto}(\alpha, \theta_{\min})$. We rescale $\theta_{\min}$ so that average productivity is fixed across values of α. Lower values of α generate more concentrated industries. Each point is one simulation. The dashed lines mark values of β^a that are 10% above or below $|\beta^b + \delta|$.

Figure 6 reports $\beta^b + \delta$ on the x-axis and β^a on the y-axis across simulations for different values of α. Panel (a) uses log growth rates estimated by OLS, while panel (b) uses standard growth rates estimated by WLS. To quantify the discrepancy between firm- and industry-level estimates, we report below each figure the percentage of simulations in which β^a and $\beta^b + \delta$ differ by more than 10%, corresponding to the points outside the dashed lines.[14]

Panel (a) shows that for low values of α, corresponding to highly concentrated industries, β^a and $\beta^b + \delta$ diverge substantially, even though SUTVA holds. Because this divergence is driven by idiosyncratic shocks affecting large firms, the sign of the difference between the coefficients can be either positive or negative. Panel (b) shows that this discrepancy disappears when using the

[14] Figure E.5 compares standard growth rates estimates using OLS and WLS and shows that using weighted least squares is necessary to achieve aggregation invariance. Relative to log growth rates with OLS, standard growth rates with OLS raise an additional issue: they can generate extreme outliers when sales in the first period are very small. Weighted least squares address this problem mechanically, since observations with small initial sales receive small weights. With OLS, however, these outliers can generate extremely large coefficients, as shown in Figure E.4. For illustration, we therefore exclude firms whose sales growth rate is in the top 1% and recompute industry aggregates accordingly.

standard growth rate with WLS. In this case, the firm-level regression assigns the same weights as the industry-level regression, so $\beta^a = \beta^b + \delta$ in all simulations and for all values of α.

4.4.3 Jensen inequality

As when disaggregating, industry-level growth is not a linear aggregation of firm-level growth rates when using log growth rates. Similarly to subsubsection 3.4.3, we define the concavity term as the difference between industry log growth and the weighted average of firm-level log growth rates:

$$C_{jt} \equiv \log\left(\sum_{i:\, j(i)=j} \omega_{ij}^{l} \exp(\Delta^{l} S_{it})\right) - \sum_{i:\, j(i)=j} \omega_{ij}^{l} \Delta^{l} S_{it}.$$

Applying the same second-order approximation as in Equation 9, we obtain that the concavity term depends on the within-industry variance of firm growth rates. Characterizing the conditions under which aggregation invariance holds, however, is more complex than in the disaggregation case because variation in treatment status within industries directly contributes to the within-industry variance of firm growth rates. To characterize the role of treatment exposure, we use Equation 2 to express firm growth rates as the sum of a treatment component and a residual.[15]

Under random assignment of treatment within industries, the covariance between treatment status and residual firm growth vanishes. Ignoring higher-order terms, the concavity term can therefore be written as

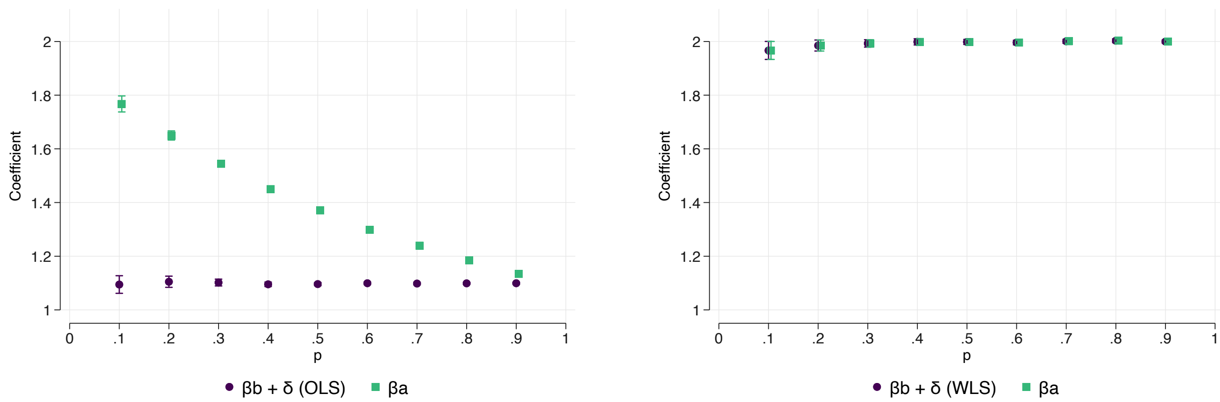
$$C_{jt} \approx \frac{1}{2}\left(\beta^2 ShareTreated_j(1 - ShareTreated_j) + \mathrm{Var}_{\omega,j}(u_{ijt})\right).$$

The first term reflects the variation due to differences in treatment status across firms within an industry. Because treatment is binary, this component depends on the share of treated firms and is largest when treatment exposure is evenly distributed within the industry. The second term reflects the residual dispersion of firm growth rates within industries. This second component is the direct counterpart of the Jensen mechanism emphasized in the disaggregation setting, where differences in within-firm dispersion of firm–market growth rates lead firm- and firm–market regressions to identify different estimands. We focus therefore on the first term in the following empirical analysis.

When will within-industry variability in treatment status lead aggregation invariance to break? Since industry log growth can be written as the sum of the weighted average of firm-level log growth rates and the concavity term, the industry-level coefficient absorbs not only the average effect of treatment on firm growth, but also the way treatment exposure mechanically affects the dispersion of firm growth rates within industries through the concavity term. Aggregation invariance under

[15] We do not include the treatment share in the computation of the concavity term since aggregation invariance is defined conditional on the absence of SUTVA violations.

Figure 7: Aggregation invariance: Jensen's inequality



(a) Log growth rate – OLS

(b) Standard growth rate – WLS

*Note.*The simulation follows Table 4. Each industry's treatment share is drawn with equal probability from either 0.01 or p_{high}, where p_{high} varies on the x-axis. Treated firms receive productivity multiplier $g = 2$. Each industry contains 10,000 firms, and there is no firm entry.

log growth therefore requires the concavity term to be orthogonal to $ShareTreated_j$.

We illustrate this mechanism using a simplified setting in which the industry-level share of treated firms can take only two values with equal probability, p_{low} and p_{high}. We set $p_{low} = 0.01$ and let p_{high} vary over the interval $[0.1, 0.9]$. Industries with p_{low} exhibit almost no within-industry variation in treatment status, since nearly all firms are untreated. By contrast, industries with p_{high} display substantially more variation in treatment status when p_{high} is below one half. Since the concavity term is increasing in $ShareTreated_j(1 - ShareTreated_j)$, it is therefore much larger in industries with p_{high} than in industries with p_{low} over this range.

As p_{high} rises above one half and approaches one, however, within-industry variation in treatment status declines again because nearly all firms become treated. The concavity term therefore decreases and eventually converges back toward the level observed in industries with p_{low}. As a result, the relationship between $ShareTreated_j$ and the concavity term is strongest for low values of p_{high} and disappears as p_{high} approaches one.

Figure 7 displays the simulation results.[16] The number of firms is fixed over time, so treatment has no effect on entry. We use a large treatment effect ($g = 2$) to make the concavity term quantitatively relevant, since it is proportional to β^2.[17] Panel (a) uses log growth rates estimated by WLS, while panel (b) uses standard growth rates estimated by WLS. As expected, the discrepancy between the firm- and industry-level estimates when using log growth rates is largest for low values of p_{high}, while aggregation invariance nearly holds when p_{high} approaches one. Panel (b) shows

[16] Figure E.6 compares log and standard growth rates estimates using WLS and shows that the standard growth rate is necessary to achieve aggregation invariance.

[17] Because industries with p_{low} contain very few treated firms, we also increase the number of firms per industry to 10,000 to ensure that treatment effects are precisely estimated.

that aggregation invariance always holds when using the standard growth rate with WLS.

4.5 Application: detection of spillovers

In this section, we show how our methodology can be used to detect SUTVA violations. Our approach allows the industry-level coefficient to be decomposed into three components: a direct treatment effect, an indirect treatment effect capturing spillovers across firms, and an effect on the entry rate. Without such a decomposition, comparing treatment effect estimates across aggregation levels can be misleading. Similar coefficients at the firm and industry levels do not necessarily imply the absence of spillovers, while differences across levels may reflect aggregation issues rather than indirect effects on incumbent firms.

We illustrate the advantages of our approach through a simple simulation in which the positive effect of treatment on entry exactly offsets negative spillovers on incumbent firms.[18] More specifically, as described in Table 4, we introduce $\bar{N} = 50$ potential entrant firms per industry. Each potential entrant enters in period t_2 with probability

$$p_{\text{entry},j} = \kappa \times ShareTreated_j,$$

where $\kappa = 0.375$. Industries with a larger share of treated firms therefore experience more entry, increasing industry growth.

At the same time, incumbent firms are exposed to a negative intensive-margin spillover. In period t_2, incumbent sales are multiplied by

$$\phi_j = 1 - \gamma \times ShareTreated_j,$$

where $\gamma = 0.25$. A larger share of treated firms therefore reduces incumbent sales growth within the industry.

Table 5 reports the simulation results. Following the same logic as subsection 3.5, we compare the ability of common practices with our approach to detect spillovers. Panel (a) reports the coefficients obtained from estimating Equation 2 (without including $ShareTreated_j$) and Equation 3 using log growth rates with OLS. The firm- and industry-level coefficients are very similar, suggesting little evidence of spillovers. Panel (b) instead applies the two-step approach presented in subsection 4.3. The coefficient on $ShareTreated_j$ in the intensive-margin regression is strongly negative, reflecting the negative spillover imposed on incumbent firms. At the same time, the coefficient on entry is positive, indicating that industries with a larger share of treated firms experience greater firm entry.

[18] An example of such a treatment could be a technological innovation that favors innovative firms over non-innovative firms while simultaneously stimulating the entry of new firms.

Table 5: Aggregation invariance: Detecting spillovers

Specification	Mean	SE
Panel A: Log growth rate – OLS		
$\hat{\beta}^a$	0.0268	0.0088
$\hat{\beta}^b$	0.0299	0.0051
Panel B: Standard growth rate – WLS		
$\hat{\beta}^a$	0.0265	0.0090
$\hat{\beta}^b + \hat{\delta} + \hat{\beta}^E$	0.0265	0.0090
$\hat{\delta}$	-0.2405	0.0089
$\hat{\beta}^b$	0.0867	0.0041
$\hat{\beta}^E$	0.1803	0.0027

Note. The simulation follows Table 4. Treated firms receive productivity multiplier $H_{treated} = 1.1$. Entry increases with industry treatment exposure ($\kappa = 0.375$). Incumbent firms are exposed to a negative spillover: their sales are multiplied by $1 - 0.25\, ShareTreated_j$. The table reports means and standard errors across 50 simulations. Panel A compares standard log-growth regressions. Panel B reports the standard-growth decomposition into the industry effect β^a, the spillover effect δ, the direct effect β^b, and the entry effect β^E.

The two effects offset each other in aggregate industry sales growth.

These results highlight a broader limitation of log-growth specifications for aggregation exercises. Because firm-level log regressions exclude entry by construction, they only capture spillovers affecting incumbent firms. When treatment simultaneously affects entry and incumbent outcomes, comparing firm- and industry-level log coefficients may therefore provide a misleading picture of the underlying economic mechanisms. In contrast, our decomposition based on standard growth rates separately identifies intensive- and extensive-margin spillovers, making it possible to recover the full aggregate effect.

5 Conclusion

Empirical researchers often move across levels of aggregation to strengthen identification or to study how micro-level treatment effects translate into aggregate outcomes. This paper shows that such comparisons are not mechanically innocuous. Even absent confounding shocks or SUTVA violations, standard specifications can generate different treatment effect estimates across aggregation levels. These differences arise because changing the level of observation changes implicit weights, alters the aggregation of nonlinear growth rates through Jensen's inequality, and changes the treatment of entry and exit.

We introduce the notions of disaggregation invariance and aggregation invariance to formalize when treatment effect estimates should be comparable across levels. For disaggregation, we show that midpoint growth rates with midpoint weights, as well as Poisson regressions in levels with appropriate fixed effects, restore invariance by aligning the estimand across baseline and disaggregated regressions. For aggregation, we show that invariance requires more than changing weights: because aggregation also changes the treatment variable, the population equations must be aligned. Our two-step approach separates incumbent-firm effects from entry effects and makes it possible to compare firm- and industry-level estimates in a way that is both statistically and economically interpretable.

The practical value of these results is that they turn comparisons across aggregation levels into more informative diagnostics. When estimates are aligned by construction, remaining differences can be interpreted as evidence of the forces researchers typically seek to uncover: confounding shocks, spillovers, or extensive-margin responses. Without such alignment, similar coefficients may be misleading, and different coefficients may reflect mechanical changes in the estimand rather than economically meaningful differences.

References

Amiti, Mary, and David Weinstein. 2018. "How Much Do Idiosyncratic Bank Shocks Affect Investment? Evidence from Matched Bank-Firm Loan Data." *Journal of Political Economy*, 126(2): 525–587.

Angrist, Joshua D, and Jörn-Steffen Pischke. 2010. "The credibility revolution in empirical economics: How better research design is taking the con out of econometrics." *Journal of economic perspectives*, 24(2): 3–30.

Arkolakis, Costas, Arnaud Costinot, and Andrés Rodríguez-Clare. 2012. "New trade models, same old gains?" *American Economic Review*, 102(1): 94–130.

Baily, Martin Neil, Charles Hulten, David Campbell, Timothy Bresnahan, and Richard E Caves. 1992. "Productivity dynamics in manufacturing plants." *Brookings papers on economic activity. Microeconomics*, 1992: 187–267.

Baker, Andrew C, David F Larcker, and Charles CY Wang. 2022. "How much should we trust staggered difference-in-differences estimates?" *Journal of Financial Economics*, 144(2): 370–395.

Beaumont, Paul, Thibault Libert, and Christophe Hurlin. 2019. "Granular Borrowers." *Working Paper.*

Beck, Thorsten, Samuel Da-Rocha-Lopes, and André F Silva. 2021. "Sharing the pain? Credit supply and real effects of bank bail-ins." *The Review of Financial Studies*, 34(4): 1747–1788.

Berg, Tobias, Markus Reisinger, and Daniel Streitz. 2021. "Spillover effects in empirical corporate finance." *Journal of Financial Economics*, 142(3): 1109–1127.

Berman, Nicolas, Philippe Martin, and Thierry Mayer. 2012. "How do different exporters react to exchange rate changes?" *The Quarterly Journal of Economics*, 127(1): 437–492.

Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan. 2004. "How much should we trust differences-in-differences estimates?" *The Quarterly Journal of Economics*, 119(1): 249–275.

Borusyak, Kirill, and Peter Hull. 2023. "Nonrandom Exposure to Exogenous Shocks." *Econometrica*, 91(6): 2155–2185.

Borusyak, Kirill, Peter Hull, and Xavier Jaravel. 2021. "Quasi-Experimental Shift-Share Research Designs." *Review of Economic Studies*, 89(1): 181–213.

Boualam, Yasser, and Clément Mazet-Sonilhac. 2025. "The Flow Approach to Credit Markets: Methodology, Measurements, and Macro Perspectives." Working paper.

Cai, Jing, and Adam Szeidl. 2024. "Indirect effects of access to finance." *American Economic Review*, 114(8): 2308–2351.

Callaway, Brantly, and Pedro Sant'Anna. 2021. "Difference-in-Differences with multiple time periods." *Journal of Econometrics*, 225(2): 200–230.

Catherine, Sylvain, Thomas Chaney, Zongbo Huang, David Sraer, and David Thesmar. 2022. "Quantifying reduced-form evidence on collateral constraints." *The Journal of Finance*, 77(4): 2143–2181.

Chemla, Gilles, and Chris Hennessy. 2026. "Discretionary Choice, Information Production, and the SUTVA." *Working paper.*

Chen, Jiafeng, and Jonathan Roth. 2024. "Logs with zeros? Some problems and solutions." *Quarterly Journal of Economics*, 139(2): 891–936.

Chodorow-Reich, Gabriel. 2014. "The employment effects of credit market disruptions: Firm-level evidence from the 2008–9 financial crisis." *The Quarterly Journal of Economics*, 129(1): 1–59.

Chodorow-Reich, Gabriel. 2019. "Geographic cross-sectional fiscal spending multipliers: What have we learned?" *American Economic Journal: Economic Policy*, 11(2): 1–34.

Cingano, Federico, Francesco Manaresi, and Enrico Sette. 2016. "Does credit crunch investment down? New evidence on the real effects of the bank-lending channel." *The Review of Financial Studies*, 29(10): 2737–2773.

Cohn, Jonathan, Zack Liu, and Malcolm Wardlaw. 2022. "Count (and count-like) data in finance." *Journal of Financial Economics*, 146(2): 529–551.

Crépon, Bruno, Esther Duflo, Marc Gurgand, Roland Rathelot, and Philippe Zamora. 2013. "Do labor market policies have displacement effects? Evidence from a clustered randomized experiment." *The quarterly journal of economics*, 128(2): 531–580.

Davis, Steven, and John Haltiwanger. 1992. "Gross Job Creation, Gross Job Destruction, and Employment Reallocation." *Quarterly Journal of Economics*, 107(3): 819–863.

Degryse, Hans, Olivier De Jonghe, Sanja Jakovljević, Klaas Mulier, and Glenn Schepens. 2019. "Identifying Credit Supply Shocks with Bank-Firm Data: Methods and Applications." *Journal of Financial Intermediation*, 40: 100813.

Di Giovanni, Julian, Andrei A Levchenko, and Isabelle Mejean. 2014. "Firms, destinations, and aggregate fluctuations." *Econometrica*, 82(4): 1303–1340.

Di Giovanni, Julian, Andrei A Levchenko, and Isabelle Mejean. 2017. "Large firms and international business cycle comovement." *American Economic Review*, 107(5): 598–602.

Eaton, Jonathan, Samuel Kortum, and Francis Kramarz. 2011. "An Anatomy of International Trade: Evidence from French Firms." *Econometrica*, 79(5): 1453–1498.

Eckert, Fabian, Teresa Fort, Peter Schott, and Natalie Yang. 2021. "Imputing Missing Values in the US Census Bureau's County Business Patterns." National Bureau of Economic Research Working Paper 26632.

Ferracci, Marc, Grégory Jolivet, and Gerard J van den Berg. 2014. "Evidence of treatment spillovers within markets." *Review of Economics and Statistics*, 96(5): 812–823.

Fitzgerald, Doireann, and Stefanie Haller. 2018. "Exporters and shocks." *Journal of International Economics*, 113: 154–171.

Fonseca, Julia, and Adrien Matray. 2024. "Financial Inclusion, Economic Development, and Inequality: Evidence from Brazil." *Journal of Financial Economics*, 156: 1–22.

Foster, Lucia, John C Haltiwanger, and Cornell John Krizan. 2001. "Aggregate productivity growth: Lessons from microeconomic evidence." In *New developments in productivity analysis*. 303–372. University of Chicago Press.

Gabaix, Xavier. 2011. "The granular origins of aggregate fluctuations." *Econometrica*, 79(3): 733–772.

Garthwaite, Craig L. 2014. "Demand spillovers, combative advertising, and celebrity endorsements." *American Economic Journal: Applied Economics*, 6(2): 76–104.

Giroud, Xavier, and Holger Mueller. 2019. "Firms' Internal Networks and Local Economic Shocks." *American Economic Review*, 109(10): 3617–3649.

Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift. 2020. "Bartik Instruments: What, When, Why, and How." *American Economic Review*, 110(8): 2586–2624.

Gormley, Todd, and David Matsa. 2013. "Common Errors: How to (and Not to) Control for Unobserved Heterogeneity." *Review of Financial Studies*, 27: 617–661.

Guren, Adam, Alisdair McKay, Emi Nakamura, and Jón Steinsson. 2021. "What do we learn from cross-regional empirical estimates in macroeconomics?" *NBER Macroeconomics Annual*, 35(1): 175–223.

Helm, Ines. 2020. "National industry trade shocks, local labour markets, and agglomeration spillovers." *The Review of Economic Studies*, 87(3): 1399–1431.

Herreno, Juan. 2023. "Aggregating the Effect of Bank Credit Supply Shocks on Firms."

Hsieh, Chang-Tai, and Peter J Klenow. 2009. "Misallocation and manufacturing TFP in China and India." *The Quarterly journal of economics*, 124(4): 1403–1448.

Huber, Kilian. 2018. "Disentangling the effects of a banking crisis: Evidence from German firms and counties." *American Economic Review*, 108(3): 868–898.

Huber, Kilian. 2023. "Estimating general equilibrium spillovers of large-scale shocks." *The Review of Financial Studies*, 36(4): 1548–1584.

Iyer, Rajkamal, José-Luis Peydró, Samuel da Rocha-Lopes, and Antoinette Schoar. 2014. "Interbank liquidity crunch and the firm credit crunch: Evidence from the 2007–2009 crisis." *The Review of Financial Studies*, 27(1): 347–372.

Jiang, Wei. 2017. "Have instrumental variables brought us closer to the truth." *Review of Corporate Finance Studies*, 6(2): 127–140.

Khwaja, Asim, and Atif Mian. 2008. "Tracing the Impact of Bank Liquidity Shocks: Evidence from an Emerging Market." *American Economic Review*, 98(4): 1413–1442.

Kline, Patrick, and Enrico Moretti. 2014. "Local economic development, agglomeration economies, and the big push: 100 years of evidence from the Tennessee Valley Authority." *The Quarterly journal of economics*, 129(1): 275–331.

Kogan, Leonid, Dimitris Papanikolaou, Amit Seru, and Noah Stoffman. 2017. "Technological innovation, resource allocation, and growth." *The Quarterly Journal of Economics*, 132(2): 665–712.

Lerche, Adrian. 2025. "Direct and indirect effects of investment tax incentives." *American Economic Review*, 115(8): 2781–2818.

Lerner, Josh, and Amit Seru. 2022. "The use and misuse of patent data: Issues for finance and beyond." *The Review of Financial Studies*, 35(6): 2667–2704.

Manski, Charles F. 1993. "Identification of endogenous social effects: The reflection problem." *The review of economic studies*, 60(3): 531–542.

Matray, Adrien, Karsten Müller, Chenzi Xu, and Poorya Kabir. 2024. "EXIM's Exit: The Real Effects of Trade Financing by Export Credit Agencies." Working Paper 32019.

McKay, Alisdair, and Christian K Wolf. 2023. "What can time-series regressions tell us about policy counterfactuals?" *Econometrica*, 91(5): 1695–1725.

Mian, Atif, and Amir Sufi. 2014. "What Explains the 2007–2009 Drop in Employment?" *Econometrica*, 82(6): 2197–2223.

Mian, Atif, Andres Pablo Sarto, and Amir Sufi. 2023. "Estimating credit multipliers."

Moulton, Brent R. 1990. "An illustration of a pitfall in estimating the effects of aggregate variables on micro units." *The Review of Economics and Statistics*, 334–338.

Nakamura, Emi, and Jón Steinsson. 2018. "Identification in macroeconomics." *Journal of Economic Perspectives*, 32(3): 59–86.

Olley, G Steven, and Ariel Pakes. 1996. "The Dynamics of Productivity in the Telecommunications Equipment Industry." *Econometrica*, 64(6): 1263–1297.

Paravisini, Daniel, Veronica Rappoport, Philipp Schnabl, and Daniel Wolfenzon. 2014. "Dissecting the Effect of Credit Supply on Trade: Evidence from Matched Credit-Export Data." *Review of Economic Studies*, 82(1): 333–359.

Petersen, Mitchell A. 2008. "Estimating standard errors in finance panel data sets: Comparing approaches." *The Review of Financial Studies*, 22(1): 435–480.

Schwert, Michael. 2018. "Bank capital and lending relationships." *The Journal of Finance*, 73(2): 787–830.

Silva, Santos, and Silvana Tenreyro. 2006. "The log of Gravity." *Review of Economics and Statistics*, 88(November): 641–658.

Sraer, David, and David Thesmar. 2023. "How to use natural experiments to estimate misallocation." *American Economic Review*, 113(4): 906–938.

Wolf, Christian K. 2023. "The missing intercept: A demand equivalence approach." *American Economic Review*, 113(8): 2232–2269.

Online Appendix for
Reconciling Treatment Effects Estimates across Levels of Aggregation

Paul Beaumont Adrien Matray Yachi Tu Chenzi Xu

A Data sources and sample construction

This appendix describes the datasets used in section 2. For each dataset, we define a baseline level and a disaggregated level, with the latter nested within the former. We also report the data source, sample period, and main construction steps used before computing the descriptive statistics. Across datasets, we impose two common restrictions. First, baseline units must be observed in at least two consecutive periods, so that the analysis focuses on continuing units. Second, observations with missing or non-positive values are dropped before aggregation.

Panjiva–Compustat. The Panjiva–Compustat data combine Panjiva vessel-level bills of lading for shipments departing the United States with Compustat firm identifiers. Shipments are linked to Compustat at the parent-firm level using a gvkey crosswalk. We use the 2013–2021 period. HS codes come from an auxiliary product file that maps each shipment's Harmonized System code to HS07, so we do not apply an additional vintage adjustment. We keep shipments for which the shipper country is the United States and shipment weight is positive. We standardize firm names, drop major non-merchant freight forwarders using name-based regular expressions, split shipment weight equally across all HS6 codes listed on a shipment, and aggregate to the gvkey $\times$ destination $\times$ HS6 product $\times$ year level. The baseline level is the firm, and the disaggregated level is firm $\times$ destination $\times$ HS6 product.

Revelio–Compustat. The Revelio–Compustat data combine employment counts from Revelio Labs with Compustat firm identifiers. Firms are linked at the gvkey level, and occupations are coded using six-digit Standard Occupational Classification codes. We use the 2014–2023 period. We keep firm $\times$ occupation $\times$ year observations with positive headcount. The baseline level is the firm, and the disaggregated level is firm $\times$ occupation.

BACI bilateral trade data. The BACI data come from CEPII and report bilateral trade values at the exporter $\times$ importer $\times$ HS6 product $\times$ year level. We use the HS07 version over the 2007–2022 period. Because CEPII publishes BACI in a fixed HS vintage, no additional product-code crosswalk is applied. We drop within-country flows, keep strictly positive trade values, and aggregate to the exporter $\times$ importer $\times$ HS6 product $\times$ year level. The sample includes eleven large advanced exporters: Australia, Denmark, Finland, France, Germany, Japan, New Zealand, Spain, Switzerland, the United Kingdom, and the United States. The baseline level is the exporter–importer pair, and the disaggregated level is exporter–importer pair $\times$ HS6 product.

U.S. County Business Patterns. The U.S. county-level employment data come from the imputation-corrected County Business Patterns panel of Eckert, Fort, Schott and Yang (2021), originally based on data from the U.S. Census Bureau. Industries are coded using the 2012 North American Industry Classification System. We use county $\times$ NAICS6 industry $\times$ year observations from 2003 to 2016. We keep observations with positive employment, drop NAICS codes that are not exactly six digits or that contain delimiter characters, and exclude two-digit sectors 11, 92, and 99, corresponding to Agriculture, Forestry, Fishing, and Hunting; Public Administration; and Unclassified industries. The baseline level is the county, and the disaggregated level is county $\times$ NAICS6 industry.

13F institutional holdings. The institutional holdings data come from Thomson Reuters S34 13F filings and are linked to monthly stock prices from CRSP using eight-digit CUSIP identifiers. The raw unit is fund manager $\times$ stock $\times$ quarter. We use the 2000–2024 period. We standardize CUSIP identifiers to eight characters, restrict the sample to common stocks with CRSP share codes 10 or 11, and drop positions below 500,000 U.S. dollars at the reporting date. To address identifier reuse, we split a manager identifier into separate manager identities whenever the manager is inactive for more than four consecutive quarters. The baseline level is the fund manager, and the disaggregated level is fund manager $\times$ stock.

Dealscan syndicated loans. The Dealscan syndicated-loan data come from WRDS Dealscan and report syndicated-loan facilities. We follow the lead-bank aggregation conventions in Schwert (2018). We use the 2000–2018 period. The activity measure is new-deal credit, dated to the year in which the deal is signed. Each facility's committed amount is split equally across the lead arrangers on the facility, since the WRDS `bankallocation` field covers only a minority of dollar volume in this vintage. We reserve allocation-based weights for a robustness check.

We keep only lead-arranger lender roles and drop non-bank institutional investors, including collateralized loan obligations, mutual funds, hedge funds, prime funds, pension funds, and other non-bank lenders, identified using the Dealscan institution-type field. We restrict the sample to borrowers in the United States and assign each borrower a four-digit SIC code using Dealscan's `primarysiccode` field, which follows the 1987 SIC vintage. For borrowers without a primary SIC code, we supplement the industry assignment using a name-based regular-expression imputation. We then aggregate facilities to the bank $\times$ borrower $\times$ year level and to the bank $\times$ borrower-SIC4 $\times$ year level. We keep banks that satisfy the activity threshold in Schwert (2018), namely at least 50 lead-arranger facilities or 5 billion U.S. dollars in lead volume over the sample window, and that are observed in at least two years.

The lead-bank identifier is time-varying. Each Dealscan lender is linked to a Compustat `gvkey` using the lender concordance in Schwert (2018), including the associated date windows. This procedure allows lender identities to follow their `gvkey` over time and absorbs most ownership changes without manual adjustments. Lenders without a match are assigned first to the Dealscan ultimate-parent identifier and then to the Dealscan company identifier. The baseline level is the bank, and the disaggregated level is bank $\times$ borrower-SIC4.

U.S. patents linked to Compustat. The firm-level patent data come from PatentsView bulk tables for granted utility patents and are linked to Compustat using the Kogan, Papanikolaou, Seru and Stoffman (2017) correspondence table and the CRSP–Compustat linking table. We use application years 2003–2013 and date patents by application year rather than grant year. The activity measure is a fractional patent count. Each patent is split equally across its distinct four-character IPC subclasses and assigned to the `gvkey` of its assignee. We keep firm $\times$ IPC4 $\times$ year cells with positive fractional patent counts and drop firms observed in fewer than two years. The baseline level is the firm, and the disaggregated level is firm $\times$ IPC4 subclass.

U.S. patents by county. The county-level patent data come from PatentsView bulk tables for granted utility patents. We use application years 2003–2013 and date patents by application year. Counties are identified using disambiguated inventor locations. The activity measure is a fractional patent count. A patent with inventors located in K distinct counties is split equally across those counties and then split equally across its distinct four-character IPC subclasses. We keep county $\times$ IPC4 $\times$ year cells with positive fractional patent counts and drop counties observed in fewer than two years. The baseline level is the county, and the disaggregated level is county $\times$ IPC4 subclass.

B Computation of simulated average treatment effects

This appendix describes how average treatment effects on firm-level sales growth are computed in our simulations.

Law of iterated expectations. Average growth rates are computed by integrating over the multiple layers of randomness in the data-generating process. Let

$$\Delta^q S_i \equiv \Delta^q(S_{i,2}, S_{i,1})$$

denote a firm-level growth transformation, where q indexes the growth-rate method. For example, $q = l$ corresponds to log growth and $q = mp$ corresponds to midpoint growth.

For a given growth-rate transformation q, the expected growth rate is obtained as

$$E[\Delta^q S_i] = E_{\theta_{i,1}}\left[E_{K_1,K_2|\theta_{i,1}}\left[E_{S_1,S_2|K_1,K_2,\theta_{i,1}}[\Delta^q(S_2, S_1)]\right]\right].$$

This decomposition highlights three conditioning steps.

(i) *Conditional on initial productivity.* We first fix an initial productivity type

$$\theta_{i,1} = \bar{\theta}_r,$$

which determines both expected market-level sales and expected market participation.

(ii) *Conditional on market participation.* Given productivity, the number of markets served in each period, $(K_{i,1}, K_{i,2})$, is drawn from the Poisson entry process. Conditional on $(K_{i,1}, K_{i,2})$, total sales are sums of $K_{i,t}$ independent exponential draws and therefore follow Gamma distributions.

(iii) *Integrating over sales draws.* Given $(K_{i,1}, K_{i,2}, \theta_{i,1})$, we integrate over the sales draws to compute the expected value of the growth transformation $\Delta^q(S_{i,2}, S_{i,1})$. For log growth, this yields an expression involving the digamma function. For midpoint growth, the expectation can be expressed using a Beta distribution representation and evaluated numerically.

Averaging over productivity. Finally, we integrate over the distribution of initial productivity types. Under the uniform discrete assumption,

$$E[\Delta^q S_i] = \frac{1}{R}\sum_{r=1}^{R} E\left[\Delta^q S_i \mid \theta_{i,1} = \bar{\theta}_r\right].$$

Average treatment effects. The simulated average treatment effect is defined as the difference in expected growth rates between treated and control firms:

$$ATE^q = E[\Delta^q S_i \mid \text{treated}] - E[\Delta^q S_i \mid \text{control}].$$

The two expectations are computed using the same iterated-expectations procedure, but under different period-2 productivity paths. For treated firms, period-2 productivity is

$$\theta_{i,2}^{T} = \theta_{i,1}(1 + g),$$

whereas for control firms it is

$$\theta_{i,2}^{C} = \theta_{i,1}.$$

C Do midpoint weights introduce bias?

To what extent is it appropriate to use midpoint weights, given that they depend on contemporaneous sales? Under firm-level random assignment, midpoint weights are completely innocuous: the disaggregated regression targets the same estimand as the baseline firm-level regression, and the latter has a causal interpretation by assumption. The fact that markets in which treated firms expand more receive greater weight is not a distortion introduced by disaggregation, but rather a feature of firm-level growth itself. Firm-level regressions mechanically place more weight on components of sales that account for a larger share of the firm's expansion.

What if treatment is not random at the firm level? The relevant case is when, conditional on market fixed effects, treatment is as good as randomly assigned at the firm–market level. Can midpoint weights make the WLS coefficient biased even though the unweighted OLS coefficient is consistent?

First, note that weights do not change the assignment mechanism. Treatment remains as good as randomly assigned conditional on market fixed effects. Weights can, however, affect the comparison made by the regression by linking the influence of an observation to the residual shock associated with that observation. To see this, consider the weighted treated–control comparison within a given market m. Its residual component is

$$\Delta_m^w = \frac{E(w_{im}u_{im} \mid Treated_i = 1, m)}{E(w_{im} \mid Treated_i = 1, m)} - \frac{E(w_{im}u_{im} \mid Treated_i = 0, m)}{E(w_{im} \mid Treated_i = 0, m)}.$$

The WLS coefficient with market fixed effects can be written as β plus a weighted average of Δ_m^w across markets, with weights determined by the within-market treatment variation and the regression weights.

With OLS, $w_{im} = 1$. Conditional random assignment then implies

$$E(u_{im} \mid Treated_i = 1, m) = E(u_{im} \mid Treated_i = 0, m) = 0,$$

so $\Delta_m^w = 0$ in each market. With midpoint weights, by contrast, the weighted residual terms can be nonzero because the weights may depend on post-treatment sales and therefore on u_{im}. The equation also makes clear, however, that nonzero weighted residuals are not by themselves a problem. Midpoint weights create bias only if the weighted average residual differs between treated and control observations within the same market.

Economically, this condition rules out cases in which midpoint weights assign different importance to idiosyncratic shocks for treated and control firms within the same market. For example, midpoint weights are problematic if treatment leads firms to expand disproportionately in markets where they receive positive residual shocks, because those shocks increase both firm–market growth and the midpoint weight attached to the observation. By contrast, if treated and control firms respond similarly to idiosyncratic shocks within markets, then any relationship between midpoint weights and residual shocks is common across treatment groups and does not bias the treated–control comparison.

Determining whether treated and control firms respond differently to residual shocks cannot be tested directly. In practice, however, a useful diagnostic is to compare midpoint weights w_{im}^{mp} with the weights implied by the standard growth rate, w_{im}^{s}, which are predetermined because they depend only on pre-treatment sales. Since standard-growth weights are defined only for markets in which the firm is present at $t-1$, we compute this comparison on the sample of pre-existing firm–market relationships. If midpoint weights are close to standard weights in this sample, then variation in w_{im}^{mp} primarily reflects predetermined differences in market exposure rather than post-treatment firm–market shocks.

Across the datasets used in section 2, midpoint weights are highly correlated with standard weights, with an average Pearson correlation of 93.8%. While this correlation does not directly test the identifying condition, it suggests that midpoint weights are driven primarily by persistent differences in firm–market size rather than by unexplained firm–market shocks.

Table C.1: Correlation between standard and midpoint weights

Dataset	Baseline	Disaggregated	Average correlation
Panjiva-Compustat	Firm	Firm × HS6 × destination	0.931
Revelio-Compustat	Firm	Firm × Occupation	0.988
US patents-Compustat	Firm	Firm × IPC4	0.938
BACI, destination-level	Exporter × importer	Exporter × importer × HS6	0.943
United States CBP	County	County × NAICS6	0.973
US patents-county	County	County × IPC4	0.918
13F (S34)	Investment manager	Investment manager × Stock	0.933
Dealscan	Bank	Bank × Borrower SIC4	0.886

D Proofs

D.1 Poisson estimator satisfies disaggregation invariance

Disaggregation invariance in Poisson regressions follows from the relationship between the first-order conditions of the disaggregated and firm-level specifications. Let

$$(\widehat{\alpha}_{im}, \widehat{\mu}_t, \widehat{\beta}_d)$$

denote a solution to the first-order conditions of the disaggregated Poisson regression in equation (8). The first-order condition for the treatment effect is

$$\sum_{i,m,t} (Treated_i \times Post_t)(S_{imt} - \widehat{S}_{imt}) = 0,$$

where

$$\widehat{S}_{imt} = \exp(\widehat{\alpha}_{im} + \widehat{\mu}_t + \widehat{\beta}_d \, Treated_i \times Post_t).$$

Because the treatment indicator is common to all markets of a given firm and period, the first-order condition can be aggregated over markets:

$$\sum_{i,t} (Treated_i \times Post_t) \left(\sum_m S_{imt} - \sum_m \widehat{S}_{imt} \right) = 0.$$

Using $S_{it} = \sum_m S_{imt}$, this becomes

$$\sum_{i,t} (Treated_i \times Post_t)(S_{it} - \widehat{S}_{it}) = 0,$$

where

$$\widehat{S}_{it} \equiv \sum_m \widehat{S}_{imt}.$$

We now show that the disaggregated estimates imply a solution to the firm-level first-order conditions. Aggregating the disaggregated fitted values gives

$$\widehat{S}_{it} = \sum_m \exp(\widehat{\alpha}_{im} + \widehat{\mu}_t + \widehat{\beta}_d \, Treated_i \times Post_t).$$

Since $\widehat{\mu}_t$ and $Treated_i \times Post_t$ do not vary across markets within a firm–period, this can be rewritten as

$$\widehat{S}_{it} = \exp(\widehat{\mu}_t + \widehat{\beta}_d \, Treated_i \times Post_t) \sum_m \exp(\widehat{\alpha}_{im}).$$

Equivalently,

$$\widehat{S}_{it} = \exp(\widehat{c}_i + \widehat{\gamma}_t + \widehat{\beta}_b \, Treated_i \times Post_t),$$

where

$$\widehat{c}_i = \log\left(\sum_m \exp(\widehat{\alpha}_{im})\right), \qquad \widehat{\gamma}_t = \widehat{\mu}_t, \qquad \widehat{\beta}_b = \widehat{\beta}_d.$$

Thus the fitted values constructed from the disaggregated regression have exactly the same exponential form as the fitted values in the baseline firm-level Poisson regression in equation (7).

It remains to verify that these constructed values satisfy the firm-level first-order conditions. For the treatment effect, the previous aggregation implies

$$\sum_{i,t}(Treated_i \times Post_t)(S_{it} - \widehat{S}_{it}) = 0,$$

which is exactly the first-order condition for β_b in the firm-level Poisson regression. The same aggregation argument applies to the fixed-effect first-order conditions. For each firm i,

$$\sum_t (S_{it} - \widehat{S}_{it}) = \sum_t \sum_m (S_{imt} - \widehat{S}_{imt}) = 0,$$

and for each period t,

$$\sum_i (S_{it} - \widehat{S}_{it}) = \sum_i \sum_m (S_{imt} - \widehat{S}_{imt}) = 0.$$

Therefore the constructed coefficient vector

$$(\widehat{c}_i, \widehat{\gamma}_t, \widehat{\beta}_b) = \left(\log\left(\sum_m \exp(\widehat{\alpha}_{im})\right), \widehat{\mu}_t, \widehat{\beta}_d\right)$$

satisfies the firm-level Poisson first-order conditions.

Under the standard identification conditions for Poisson pseudo-maximum likelihood, the fitted values and the identified treatment coefficient solving the firm-level first-order conditions are unique, up to the usual normalization of fixed effects. Hence the firm-level estimate must satisfy

$$\widehat{\beta}_b = \widehat{\beta}_d.$$

This establishes disaggregation invariance for the Poisson specification.

D.2 Midpoint growth rate with midpoint weights satisfy disaggregation invariance

Consider the baseline WLS regression

$$\Delta^{mp} S_{it} = \alpha^b + \beta^b Treated_i + \epsilon_{it},$$

and the disaggregated WLS regression

$$\Delta^{mp} S_{imt} = \alpha^d + \beta^d Treated_i + \epsilon_{imt}.$$

The disaggregated regression is estimated using midpoint weights. Let

$$z_i = \begin{pmatrix} 1 \\ Treated_i \end{pmatrix}$$

denote the regressor vector. Since treatment is defined at the firm level, z_i is common to all markets m of firm i.

The disaggregated WLS first-order condition is

$$\sum_i \sum_{m \in m(i)} w_{im}^{mp} z_i \left(\Delta^{mp} S_{imt} - z_i' \widehat{\theta}^d \right) = 0,$$

where

$$\widehat{\theta}^d = \begin{pmatrix} \widehat{\alpha}^d \\ \widehat{\beta}^d \end{pmatrix}.$$

Because z_i does not vary across markets within firm, this can be rewritten as

$$\sum_i z_i \left[\sum_{m \in m(i)} w_{im}^{mp} \left(\Delta^{mp} S_{imt} - z_i' \widehat{\theta}^d \right) \right] = 0.$$

Using the normalization of the weights and the midpoint-growth aggregation property,

$$\sum_{m \in m(i)} w_{im}^{mp} \Delta^{mp} S_{imt} = \Delta^{mp} S_{it},$$

we obtain

$$\sum_i z_i \left(\Delta^{mp} S_{it} - z_i' \widehat{\theta}^d \right) = 0.$$

This is exactly the first-order condition of the baseline OLS regression,

$$\sum_i z_i \left(\Delta^{mp} S_{it} - z_i' \widehat{\theta}^b \right) = 0,$$

where

$$\widehat{\theta}^b = \begin{pmatrix} \widehat{\alpha}^b \\ \widehat{\beta}^b \end{pmatrix}.$$

Therefore, the disaggregated estimates satisfy the baseline first-order conditions. Under the usual full-rank condition for the weighted regressor matrix, the OLS solution is unique. Hence,

$$\widehat{\theta}^d = \widehat{\theta}^b.$$

Equivalently,

$$\widehat{\alpha}^d = \widehat{\alpha}^b, \qquad \widehat{\beta}^d = \widehat{\beta}^b.$$

D.3 Inference under disaggregation invariance

We now show that the same aggregation logic implies equality of the cluster-robust variance matrix when clustering is performed at the baseline level. Consider a WLS regression with coefficient vector θ. The cluster-robust variance estimator can be written in sandwich form as

$$\widehat{V}_{cl}(\widehat{\theta}) = \widehat{A}^{-1}\widehat{B}\widehat{A}^{-1},$$

where the bread is

$$\widehat{A} = \sum_{\ell} w_{\ell} z_{\ell} z_{\ell}',$$

and the meat is

$$\widehat{B} = \sum_{c} \widehat{g}_c \widehat{g}_c', \qquad \widehat{g}_c = \sum_{\ell \in c} w_{\ell} z_{\ell} \widehat{\epsilon}_{\ell}.$$

Here, ℓ indexes observations, c indexes clusters, z_{ℓ} is the regressor vector, w_{ℓ} is the WLS weight, and $\widehat{\epsilon}_{\ell}$ is the estimated residual.

We compare the sandwich components in the baseline and disaggregated regressions. Because clustering is performed at the firm level, the cluster is i. The firm-level cluster score in the disaggregated regression is

$$\widehat{g}_i^d = \sum_{m \in m(i)} w_{im}^{mp} z_i \widehat{\epsilon}_{imt},$$

where

$$\widehat{\epsilon}_{imt} = \Delta^{mp} S_{imt} - z_i' \widehat{\theta}^d.$$

Since z_i is constant across markets within firm,

$$\begin{aligned} \widehat{g}_i^d &= z_i \sum_{m \in m(i)} w_{im}^{mp} \left(\Delta^{mp} S_{imt} - z_i' \widehat{\theta}^d \right) \\ &= z_i \left[\sum_{m \in m(i)} w_{im}^{mp} \Delta^{mp} S_{imt} - z_i' \widehat{\theta}^d \sum_{m \in m(i)} w_{im}^{mp} \right]. \end{aligned}$$

Using the midpoint-growth aggregation property and the normalization of the weights, this becomes

$$\widehat{g}_i^d = z_i \left(\Delta^{mp} S_{it} - z_i' \widehat{\theta}^d \right).$$

By the coefficient invariance result above, $\widehat{\theta}^d = \widehat{\theta}^b$. Therefore,

$$\widehat{g}_i^d = z_i \left(\Delta^{mp} S_{it} - z_i' \widehat{\theta}^b \right) = \widehat{g}_i^b,$$

where $\widehat{g}_i^b$ is the firm-level cluster score in the baseline regression. Hence,

$$\widehat{g}_i^d = \widehat{g}_i^b$$

for every firm i. It follows that the meat is identical across the two regressions:

$$\widehat{B}^d = \sum_i \widehat{g}_i^d \widehat{g}_i^{d\prime} = \sum_i \widehat{g}_i^b \widehat{g}_i^{b\prime} = \widehat{B}^b.$$

The bread is also identical. In the disaggregated regression,

$$\widehat{A}^d = \sum_i \sum_{m \in m(i)} w_{im}^{mp} z_i z_i'.$$

Since z_i does not vary across markets within firm and the weights sum to one within firm,

$$\widehat{A}^d = \sum_i z_i z_i' \sum_{m \in m(i)} w_{im}^{mp} = \sum_i z_i z_i' = \widehat{A}^b.$$

Thus,

$$\widehat{A}^d = \widehat{A}^b \qquad \text{and} \qquad \widehat{B}^d = \widehat{B}^b.$$

Therefore,

$$\widehat{V}_{cl(i)}^d(\widehat{\theta}) = \left(\widehat{A}^d\right)^{-1} \widehat{B}^d \left(\widehat{A}^d\right)^{-1} = \left(\widehat{A}^b\right)^{-1} \widehat{B}^b \left(\widehat{A}^b\right)^{-1} = \widehat{V}_{cl(i)}^b(\widehat{\theta}).$$

The equality holds for the full variance-covariance matrix. Therefore, it also holds for the variance of the treatment coefficient:

$$\widehat{V}_{cl(i)}^d(\widehat{\beta}) = \widehat{V}_{cl(i)}^b(\widehat{\beta}),$$

and hence

$$\widehat{SE}_{cl(i)}^d(\widehat{\beta}) = \widehat{SE}_{cl(i)}^b(\widehat{\beta}).$$

D.4 Aggregation invariance

Without loss of generality, we assume no entry at the firm level. For simplicity, we use the notation ST_j for $ShareTreated_j$. This variable is defined as the pre-period-sales-weighted treatment share in industry j:

$$ST_j = \sum_{i:\, j(i)=j} \omega_{ij}^s Treated_i,$$

where ω_{ij}^s denotes firm i's pre-period sales share within industry j, with

$$\sum_{i:\, j(i)=j} \omega_{ij}^s = 1.$$

By construction,

$$\sum_{i:\, j(i)=j} \omega_{ij}^s (Treated_i - ST_j) = 0.$$

The augmented firm-level regression is

$$\Delta^s S_{it} = \alpha + \beta^b Treated_i + \delta ST_j + \epsilon_{it}.$$

Using the decomposition

$$Treated_i = (Treated_i - ST_j) + ST_j,$$

we can rewrite the regression as

$$\Delta^s S_{it} = \alpha + \beta^b (Treated_i - ST_j) + (\beta^b + \delta) ST_j + \epsilon_{it}.$$

We now show that the intercept and the coefficient on ST_j in this reparameterized firm-level regression coincide with the intercept and slope in the industry-level regression. Throughout the proof, expectations and projections are taken with respect to the same weighted measure, where firm i's within-industry weight is ω^s_{ij}.

Let

$$\theta^f = \begin{pmatrix} \alpha \\ \beta^b + \delta \end{pmatrix}, \qquad x_j = \begin{pmatrix} 1 \\ ST_j \end{pmatrix}.$$

The reparameterized firm-level regression also includes the within-industry component $Treated_i - ST_j$. The first-order conditions for the intercept and for ST_j are

$$E\Big[x_j \left(\Delta^s S_{it} - \alpha - \beta^b (Treated_i - ST_j) - (\beta^b + \delta) ST_j\right)\Big] = 0.$$

Equivalently,

$$E\Big[x_j \left(\Delta^s S_{it} - x_j' \theta^f - \beta^b (Treated_i - ST_j)\right)\Big] = 0.$$

We aggregate this moment condition within each industry. Since x_j is constant within industry and

$$\sum_{i:\, j(i)=j} \omega^s_{ij} (Treated_i - ST_j) = 0,$$

the within-industry component drops out:

$$\sum_{i:\, j(i)=j} \omega^s_{ij} x_j \beta^b (Treated_i - ST_j) = x_j \beta^b \sum_{i:\, j(i)=j} \omega^s_{ij} (Treated_i - ST_j) = 0.$$

Therefore, the firm-level moment condition reduces to

$$E\left[x_j \left(\sum_{i:\, j(i)=j} \omega^s_{ij} \Delta^s S_{it} - x_j' \theta^f\right)\right] = 0.$$

Using the linear aggregation of standard growth rates,

$$\sum_{i:\, j(i)=j} \omega^s_{ij} \Delta^s S_{it} = \Delta^s S_{jt},$$

we obtain

$$E\Big[x_j \left(\Delta^s S_{jt} - x_j' \theta^f\right)\Big] = 0.$$

This is exactly the population first-order condition for the industry-level regression

$$\Delta^s S_{jt} = \alpha^a + \beta^a ST_j + e_{jt},$$

whose coefficient vector

$$\theta^a = \begin{pmatrix} \alpha^a \\ \beta^a \end{pmatrix}$$

satisfies

$$E\big[x_j \left(\Delta^s S_{jt} - x_j' \theta^a\right)\big] = 0.$$

Under the usual full-rank condition for the industry-level regressor matrix, the solution is unique. Hence,

$$\theta^a = \theta^f.$$

Therefore,

$$\alpha^a = \alpha \qquad \text{and} \qquad \beta^a = \beta^b + \delta.$$

D.5 Inference under aggregation invariance

This section shows that, under the specification presented in subsection 4.3, the standard error around the firm-level linear combination $\widehat{\beta}^b + \widehat{\delta}$ is the same as the standard error around the industry-level coefficient $\widehat{\beta}^a$ when standard errors are clustered at the industry level. Without loss of generality, we assume no entry for the sake of simplicity. Let

$$ST_j \equiv Share\,Treated_j.$$

Consider the firm-level WLS regression

$$\Delta^s S_{it} = \alpha^b + \beta^b Treated_i + \delta ST_j + u_{it}^b,$$

estimated with weights ω_{ij}^s, with standard errors clustered at the industry level. We want inference on

$$\theta \equiv \beta^b + \delta.$$

As in subsection D.4, we rewrite the regression as

$$\Delta^s S_{it} = \alpha^b + \beta^b \left(Treated_i - ST_j\right) + \theta ST_j + u_{it}^b.$$

The industry-level regression is

$$\Delta^s S_{jt} = \alpha^a + \beta^a ST_j + u_{jt}^a.$$

Equality of the cluster-level scores. Focus on the score contribution for the aggregated block consisting of the constant and the coefficient on ST_j. In the reparameterized firm-level regression, this coefficient is θ; in the industry-level regression, it is β^a. Let

$$Z_j = \begin{pmatrix} 1 \\ ST_j \end{pmatrix}.$$

At the firm level, the industry-clustered score contribution for this block is

$$\widehat{g}_j^b = \sum_{i:j(i)=j} \omega_{ij}^s Z_j \widehat{u}_{it}^b,$$

where

$$\widehat{u}_{it}^b = \Delta^s S_{it} - \widehat{\alpha}^b - \widehat{\beta}^b \left(Treated_i - ST_j\right) - \widehat{\theta} ST_j.$$

Since Z_j is constant within industry, substituting the residual into the score gives

$$\widehat{g}_j^b = Z_j \left[\sum_{i:j(i)=j} \omega_{ij}^s \Delta^s S_{it} - \widehat{\alpha}^b \sum_{i:j(i)=j} \omega_{ij}^s - \widehat{\beta}^b \sum_{i:j(i)=j} \omega_{ij}^s \left(Treated_i - ST_j\right) - \widehat{\theta} ST_j \sum_{i:j(i)=j} \omega_{ij}^s \right].$$

Using the aggregation properties of the standard growth rate,

$$\sum_{i:j(i)=j} \omega_{ij}^s \Delta^s S_{it} = \Delta^s S_{jt}, \qquad \sum_{i:j(i)=j} \omega_{ij}^s = 1,$$

and

$$\sum_{i:j(i)=j} \omega_{ij}^s \left(Treated_i - ST_j\right) = 0,$$

we obtain

$$\widehat{g}_j^b = Z_j \left(\Delta^s S_{jt} - \widehat{\alpha}^b - \widehat{\theta} ST_j \right).$$

Using the fact that the specification delivers aggregation invariance,

$$\widehat{\alpha}^b = \widehat{\alpha}^a, \qquad \widehat{\theta} = \widehat{\beta}^a,$$

so

$$\widehat{g}_j^b = Z_j \left(\Delta^s S_{jt} - \widehat{\alpha}^a - \widehat{\beta}^a ST_j \right).$$

The right-hand side is exactly the industry-level score contribution,

$$\widehat{g}_j^a = Z_j \widehat{u}_{jt}^a = Z_j \left(\Delta^s S_{jt} - \widehat{\alpha}^a - \widehat{\beta}^a ST_j \right).$$

Therefore,

$$\widehat{g}_j^b = \widehat{g}_j^a$$

for every industry j. It follows that the meat of the industry-clustered sandwich estimator is identical for the aggregated block:

$$\widehat{B}^b = \sum_j \widehat{g}_j^b \widehat{g}_j^{b\prime} = \sum_j \widehat{g}_j^a \widehat{g}_j^{a\prime} = \widehat{B}^a.$$

Equality of the bread. The bread is also identical for the aggregated block. In the firm-level WLS regression, the weighted second-moment matrix for the constant and ST_j is

$$\sum_j \sum_{i:j(i)=j} \omega_{ij}^s Z_j Z_j'.$$

Since Z_j is constant within industry and the weights sum to one within industry,

$$\sum_j \sum_{i:j(i)=j} \omega_{ij}^s Z_j Z_j' = \sum_j Z_j Z_j' \sum_{i:j(i)=j} \omega_{ij}^s = \sum_j Z_j Z_j'.$$

This is exactly the corresponding industry-level bread for the constant and ST_j.

The additional firm-level regressor,

$$Treated_i - ST_j,$$

does not affect the bread for the aggregated block because it aggregates to zero within industry:

$$\sum_{i:j(i)=j} \omega_{ij}^s \left(Treated_i - ST_j\right) = 0.$$

Thus, the weighted cross-product between the aggregated block and $Treated_i - ST_j$ is

$$\sum_j \sum_{i:j(i)=j} \omega_{ij}^s Z_j \left(Treated_i - ST_j\right) = \sum_j Z_j \sum_{i:j(i)=j} \omega_{ij}^s \left(Treated_i - ST_j\right) = 0.$$

Hence, the bread for the aggregated block is the same at the firm and industry levels:

$$\widehat{A}^b = \widehat{A}^a.$$

Since both the bread and the meat are identical for the aggregated block,

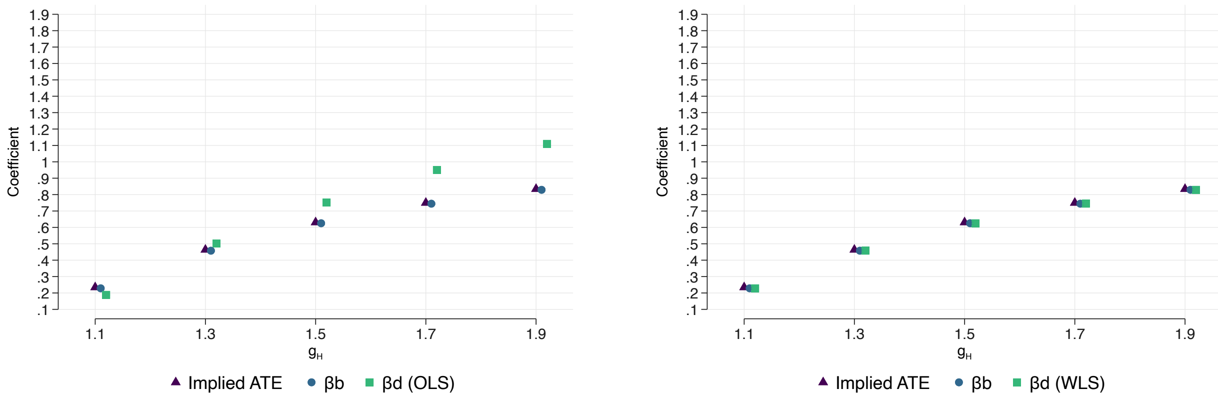
$$\widehat{V}_{cl(j)}^b = \left(\widehat{A}^b\right)^{-1} \widehat{B}^b \left(\widehat{A}^b\right)^{-1} = \left(\widehat{A}^a\right)^{-1} \widehat{B}^a \left(\widehat{A}^a\right)^{-1} = \widehat{V}_{cl(j)}^a.$$

The variance of $\widehat{\theta} = \widehat{\beta}^b + \widehat{\delta}$ is the $(2,2)$ element of this matrix, which is identical to the variance of $\widehat{\beta}^a$. Therefore,

$$\widehat{SE}_{cl(j)}^b \left(\widehat{\beta}^b + \widehat{\delta}\right) = \widehat{SE}_{cl(j)}^a \left(\widehat{\beta}^a\right).$$

E Additional results

Figure E.1: Disaggregation invariance: Implicit weights

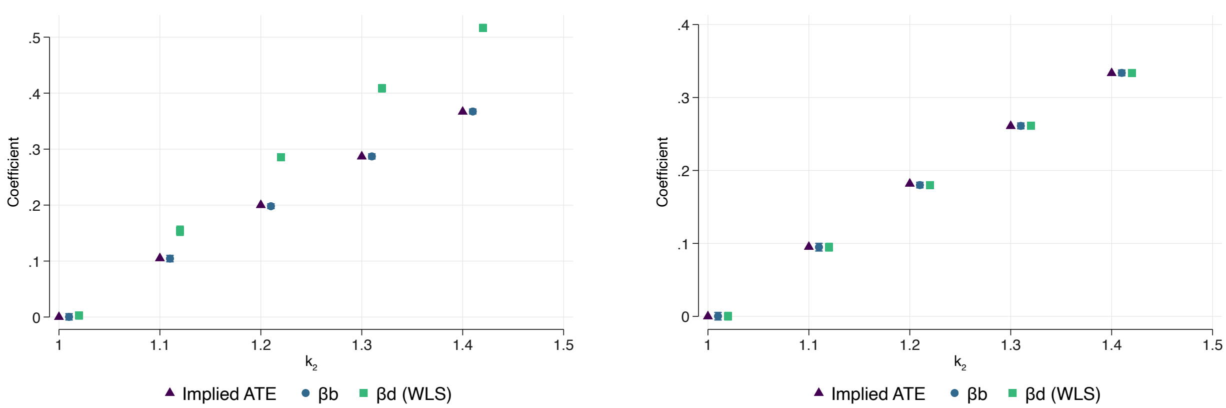


(a) Midpoint growth rate – OLS

(b) Midpoint growth rate – WLS

Note. The simulation follows Table 1. Firms draw low or high initial productivity with equal probability. Low-productivity firms have $\theta_L = 0.1$, and high-productivity firms have $\theta_H = 3$. Pre-period productivity determines the number of markets, with $\eta = 3$. The number of markets is then held fixed across periods. Treated low-productivity firms receive productivity multiplier $g_L = 1.1$. Treated high-productivity firms receive the multiplier g_H shown on the x-axis. Control firms receive no productivity change.

Figure E.2: Disaggregation invariance: Jensen's inequality

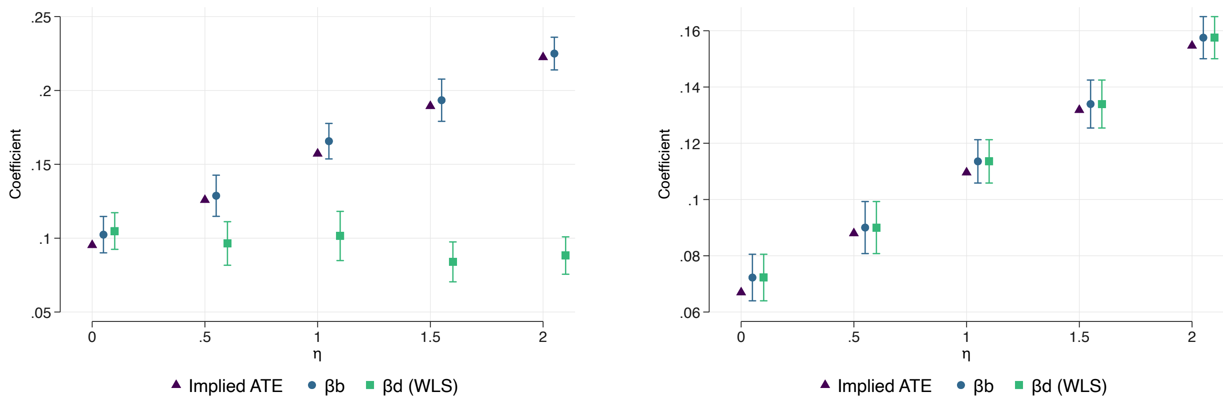


(a) Log growth rate – WLS

(b) Midpoint growth rate – WLS

Note. The simulation follows Table 1. All firms operate in five markets in both periods. Firm–market sales are independent Gamma draws with shape k_t and rate λ_t. Control firms keep the same distribution in both periods. Treated firms start from the same distribution as controls, but in period 2 their sales are drawn with shape parameter k_2, shown on the x-axis. The remaining Gamma parameters are fixed at $k_1 = \lambda_1 = 1$.

Figure E.3: Disaggregation invariance: entry and exit

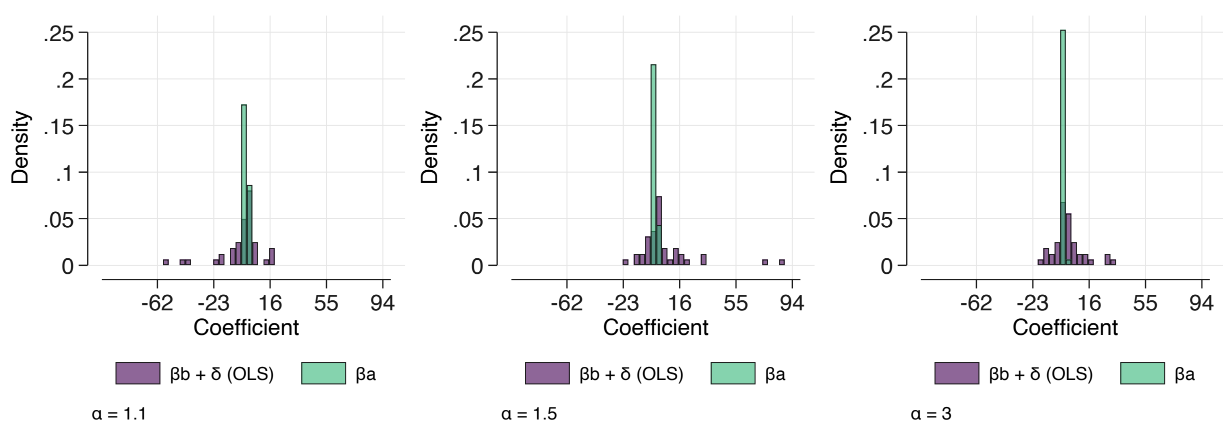


(a) Log growth rate – WLS

(b) Midpoint growth rate – WLS

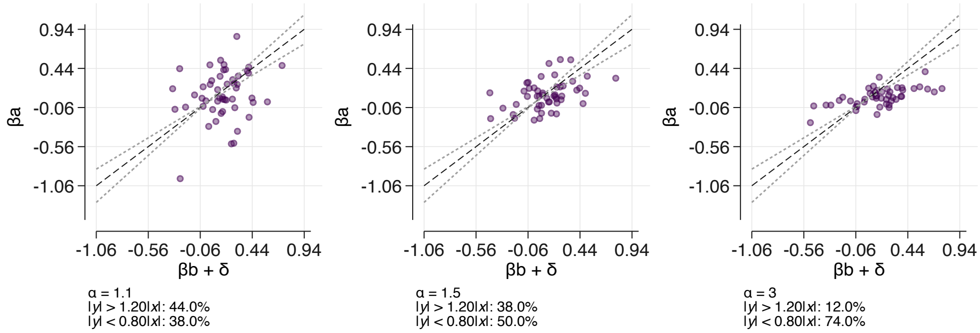
Note. The simulation follows Table 1. The parameter η, which governs how strongly market participation responds to productivity, varies on the x-axis.

Figure E.4: Aggregation: Outliers in implicit weights simulation

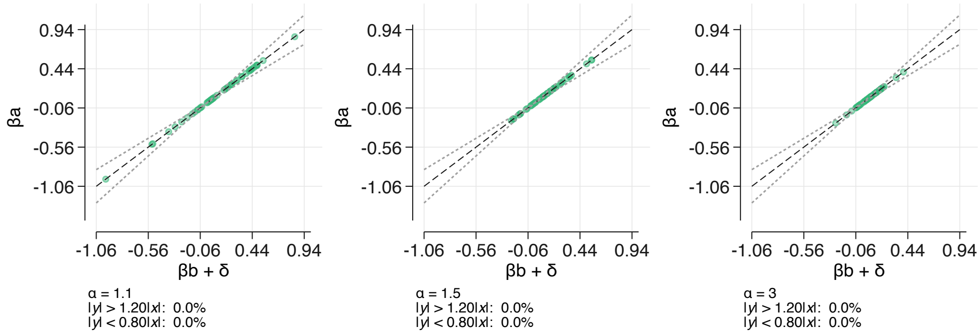


Note. Each figure shows the distribution of the coefficients obtained in 50 simulations for different values of α. Firm- and industry-level regressions use OLS regressions with standard growth rates.

Figure E.5: Aggregation invariance: Implicit weights



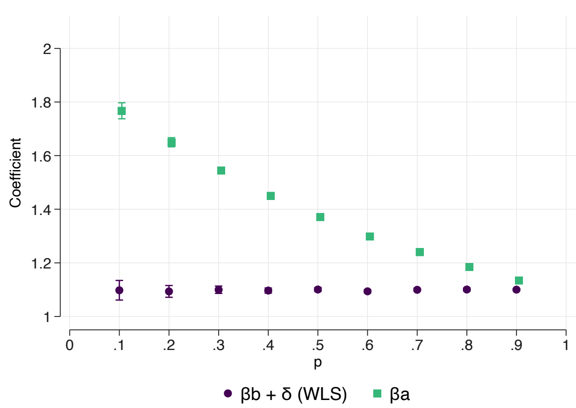
(a) Standard growth rate – OLS



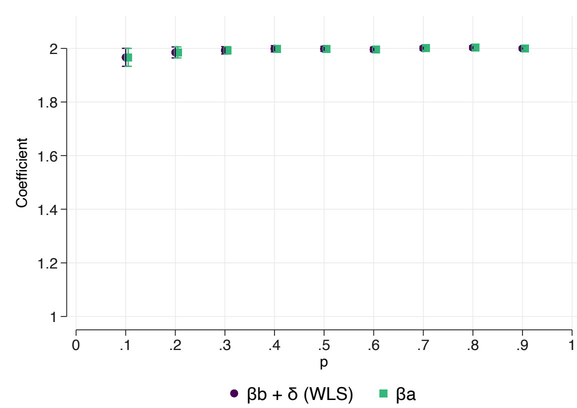
(b) Standard growth rate – WLS

Note. The figure plots $\beta^b + \delta$ on the x-axis and β^a on the y-axis across simulations for different values of α. The simulation follows Table 4. Initial firm productivity follows a Pareto distribution, $\theta_{i1} \sim \text{Pareto}(\alpha, \theta_{\min})$. We rescale $\theta_{\min}$ so that average productivity is fixed across values of α. Lower values of α generate more concentrated industries. Each point is one simulation. The dashed lines mark values of β^a that are 10% above or below $|\beta^b + \delta|$. In both panels, we exclude firms in the top 1% of sales growth rate.

Figure E.6: Aggregation invariance: Jensen's inequality



(a) Log growth rate – WLS



(b) Standard growth rate – WLS

Note. The simulation follows Table 4. Each industry's treatment share is drawn with equal probability from either 0.01 or p_{high}, where p_{high} varies on the x-axis. Treated firms receive productivity multiplier $g = 3$. Each industry contains 10,000 firms, and there is no firm entry.